

THE FOURIER RING IS NOT THE ALGORITHM: CAUSALLY DISSOCIATING LOGIT-FOURIER COMPUTA- TION FROM EMBEDDING GEOMETRY IN GROKING

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ABSTRACT

The visible Fourier ring in grokked modular-arithmetic transformers is often treated as the geometric face of the learned algorithm. We show a sharper separation: output-logit Fourier subspaces become causally load-bearing before behavioral generalization, while the embedding ring consolidates afterward. In the same runs, we estimate three onsets: logit-Fourier alignment τ_{circuit} , generalization τ_{gen} , and embedding-Fourier alignment τ_F . The strongest evidence is a time-resolved intervention: in the interval before the embedding ring consolidates, ablating the dominant logit-Fourier harmonics reduces test accuracy by 0.58–0.72, whereas matched-energy random and off-frequency Fourier controls are essentially inert. The timing evidence agrees with this causal picture: the proxy-level ordering $\tau_{\text{circuit}} < \tau_{\text{gen}} < \tau_F$ holds in 48/54 primary runs, and the broader embedding-lag claim $G < F$ holds in 236/260 Grokking runs across task, prime, dense-grid, hardware, and slow-grokking checks. These measurements refine the common claim that embedding structure emerges with generalization: the ring is better read as a post-generalization consolidation signal than as the structure through which the computation first becomes load-bearing.

1 INTRODUCTION

Grokking, the abrupt onset of generalization long after training accuracy saturates, has become a canonical testbed for studying delayed representation learning in neural networks (Power et al., 2022). Subsequent work has both extended grokking far beyond toy modular arithmetic, documenting it on broader architectures and datasets (Liu et al., 2023; Humayun et al., 2024; Miller et al., 2024), and developed mechanistic and theoretical accounts of *when* and *why* the transition occurs (Nanda et al., 2023; Varma et al., 2023; Lyu et al., 2024; Rubin et al., 2024; Kumar et al., 2024; Mohamadi et al., 2024). On modular arithmetic tasks, Nanda et al. (2023) showed that grokked transformers implement a Fourier circuit: the network uses trigonometric identities over a small set of task-relevant harmonics to compute modular addition. As a consequence, the p token embeddings form a near-circular ring in PCA space, with embedding i at angle $2\pi i/p$. That visualization is compelling, but it can also conflate two questions: whether Fourier structure is causally used for the computation, and whether the *embedding table itself* has already settled into a Fourier geometry.

Nanda et al. (2023) also introduced mechanism-level *progress measures*, such as the Fourier-squared loss, and found that these measures improve during a circuit-formation phase that *precedes* the generalization jump. This raises a natural question about a related but distinct quantity: the token *embedding geometry* itself. Does embedding-level Fourier alignment also consolidate before generalization, or does it lag behind?

Answering this requires measuring the relative timing of two events:

- τ_{gen} : generalization onset, the first checkpoint where test accuracy ≥ 0.9 and remains there for three consecutive evaluations;
- τ_F : embedding-geometry onset, the step at which token embeddings first show significantly Fourier-aligned structure above a permutation-null baseline (BIC changepoint on null-corrected Fourier R^2).

Prior work has not directly measured $\Delta\tau = \tau_F - \tau_{\text{gen}}$. Nanda et al. (2023) studied mechanism-level dynamics but did not separately track embedding geometry as an onset. Liu et al. (2022) observed that grokking “coincides with the emergence of structure in the embeddings”, a qualitative claim that our within-run measurement refines to a quantitative lag: embedding-Fourier consolidation typically follows behavioral generalization by a median of $\sim 6,000$ steps. Our work closes this gap by separating *when* the load-bearing Fourier computation becomes necessary from *when* the visible Fourier geometry appears.

Contributions. Our contribution is a timing- and intervention-based separation between the load-bearing Fourier computation and the visible Fourier geometry in modular-arithmetic grokking.

(i) The visible embedding Fourier ring is not where the load-bearing Fourier computation lives at the moment of generalization. We provide time-resolved causal evidence: before the embedding ring consolidates, removing the dominant logit-Fourier harmonics sharply damages test accuracy, while matched-energy random removals and off-frequency Fourier removals are essentially inert. Top- K and frequency-selectivity controls verify that the effect is localized to high-power low-order logit-Fourier structure rather than arbitrary low-rank damage.

(ii) A three-onset timing framework operationalizes the dissociation. We estimate logit-Fourier alignment (τ_{circuit}), behavioral generalization (τ_{gen}), and embedding-Fourier consolidation (τ_F) in the same runs via BIC changepoint detection on null-corrected Fourier alignment. The proxy-level ordering $\tau_{\text{circuit}} < \tau_{\text{gen}} < \tau_F$ holds in 48/54 primary runs and 44/55 in a cross-hardware replication, refining Nanda et al. (2023)’s circuit-formation account by isolating the visible embedding ring as a distinct, later event. We explicitly treat this as a proxy-level timing claim: under coarser loss-based progress measures, the three-stage rate drops substantially (Appendix J.5, E3).

(iii) The embedding-ring lag is robust across regimes. $G < F$ holds in 236/260=90.8% of Grokking runs across two tasks, two primes, a dense 2D (lr, wd) grid, and hardware. In particular, a targeted slow-grokking sweep falsifies the alternative explanation that the lag is merely a speed artefact, and detector sensitivity studies show that cells with clear margins do not flip ordering under the tested specifications.

2 METHODS

2.1 TASK AND MODEL

We train a decoder-only transformer (Vaswani et al., 2017) on $(a + b) \bmod 53$. All $p^2 = 2,809$ input pairs are generated; 30% (843 pairs) form the training set. The model uses $d_{\text{model}} = 256$, 4 attention heads, 2 layers, feed-forward dimension 1024, no dropout. The embedding layer has fixed learning rate $\eta_{\text{emb}} = 10^{-3}$ and weight decay 0; only decoder parameters are swept. We use AdamW (Loshchilov & Hutter, 2019) with a 10-step linear warmup followed by constant learning rate, and gradient clipping at norm 1.0.

2.2 METRICS

Generalization onset τ_{gen} . Test accuracy is evaluated every 500 steps. τ_{gen} is the first checkpoint where test accuracy ≥ 0.9 for three consecutive evaluations (1,500 steps of sustained performance).

Fourier alignment score $F(t)$. Let $E_t \in \mathbb{R}^{p \times d}$ be the (mean-row-centered) token embedding matrix at step t . We compute the fraction of variance explained by the best single Fourier harmonic:

$$F_{\text{raw}}(t) = \max_{k=1}^{\lfloor p/2 \rfloor} \left(1 - \frac{\|E_t - Q_k Q_k^\top E_t\|_F^2}{\|E_t\|_F^2} \right), \quad (1)$$

where Q_k is an orthonormal basis for $[\cos(2\pi ki/p), \sin(2\pi ki/p)]_{i=0}^{p-1}$.

Null correction. With $p = 53$ and $d = 256$, any embedding matrix achieves non-trivial F_{raw} by chance. Unless otherwise stated, the canonical detector refreshes the permutation null at every 500-step log point, matching the metric-logging cadence; for sensitivity analysis we also evaluate

refresh intervals of 2,500 and 5,000 steps (Section J.4). At each refresh step we estimate the 95th percentile of F_{raw} under 100 random token-index permutations of the *current* embedding matrix:

$$F_{\text{null},p95}(t) = \text{Pct}_{95}\left(\{F_{\text{raw}}(E_t[\sigma]) \mid \sigma \in \mathcal{S}^{(100)}\}\right), \quad (2)$$

and set $F_{\text{corr}}(t) = \max(0, F_{\text{raw}}(t) - F_{\text{null},p95}(t))$.

Fourier geometry onset τ_F . We apply a BIC-selected 1-break segmented regression (Schwarz, 1978) to $F_{\text{corr}}(t)$ on a log-time scale after EMA smoothing ($\alpha=0.15$). The breakpoint is accepted only if the 4-parameter model strictly improves BIC over the 2-parameter no-break baseline and the post-break slope exceeds 1% of the total value range; τ_F is the accepted breakpoint, or “not detected” otherwise. Log-time motivation, BIC residual assumptions, and the null-refresh artefact check are deferred to Appendix L.

Logit Fourier alignment score $F_L(t)$ and circuit onset τ_{circuit} . Let $\mathbf{L}_t \in \mathbb{R}^{p^2 \times p}$ be the full logit matrix at step t , where row $(ap + b)$ is the output logit vector for input pair (a, b) . Let $s_{ab} = (a + b) \bmod p$ be the target class. Column-centering $\mathbf{L}_t^c = \mathbf{L}_t - \bar{\mathbf{L}}_t$, we compute:

$$F_L(t) = \max_{k=1}^{\lfloor p/2 \rfloor} \left(1 - \frac{\|\mathbf{L}_t^c - Q_k^{(s)} Q_k^{(s)\top} \mathbf{L}_t^c\|_F^2}{\|\mathbf{L}_t^c\|_F^2} \right), \quad (3)$$

where $Q_k^{(s)}$ is an orthonormal basis for $[\cos(2\pi k s_{ab}/p), \sin(2\pi k s_{ab}/p)]$ indexed over all p^2 input pairs. This measures the fraction of total logit variance explained by the best single Fourier harmonic of the answer class, analogous to Nanda et al.’s restricted logit loss. A permutation-null-corrected version F_L^{corr} is computed identically to F_{corr} (permuting the s_{ab} labels). τ_{circuit} is the BIC changepoint of F_L^{corr} , estimated with the same procedure as τ_F .

Ordering labels. Given $(\tau_{\text{gen}}, \tau_F)$, each run is labeled: **G<F** ($\tau_{\text{gen}} < \tau_F$); **F<G** ($\tau_{\text{gen}} > \tau_F$); **coincident** ($|\Delta\tau| \leq 500$); **F_only** (τ_F detected, τ_{gen} not); or **none** (neither detected). Three-way ordering among $(\tau_{\text{circuit}}, \tau_{\text{gen}}, \tau_F)$ is denoted by the corresponding inequality string (e.g., **C<G<F**).

2.3 EXPERIMENTAL DESIGN

A run is *Grokking* if $\tau_{\text{gen}} - \tau_{\text{train}} \geq 2,000$ steps; *Memorization* if only τ_{train} is detected; *Comprehension* otherwise. Six experimental stages sweep the (lr, wd) phase diagram and stress-test ordering across tasks/primes: Stages 2–3 are 1D sweeps at canonical operating points (60 runs); Stage 4 retrains all Stage 2/3 cells with the F_L metric to recover τ_{circuit} ; Stage 5A is modular multiplication via discrete-log reindexing (30 runs); Stage 5B moves to $p=97$ (30 runs); Stage 6 is a dense 7×7 (lr, wd) grid (147 runs). All cells use seeds $\{42, 7, 2025\}$, 50,000 training steps (auto-extended to 80,000 if $\text{train_acc} > 0.9$ before τ_{gen} detection), and the canonical detector configuration above. Full hyperparameter grids, auto-extension rules, and the discrete-log reindexing procedure for Stage 5A are in Appendix B.

Code, training configurations, and per-run metric trajectories will be released at <https://github.com/ANONYMISED/three-stage-grokking> (link anonymised for review). Exact commit hashes, random seeds, and package versions used to produce each figure and table are listed in Appendix E.

3 RESULTS

Figure 1 provides intuition for the two key events and their ordering before we present aggregate statistics.

3.1 STAGES 2 AND 3: G<F ACROSS THE 1D SWEEPS

Stages 2 and 3 sweep weight decay (lr fixed at 1.6×10^{-3}) and learning rate (wd fixed at 2.5) over the Grokking phase of the (lr, wd) phase diagram. Of the 60 cells: **Stage 2:** 30/30 Grokking, **29/30**

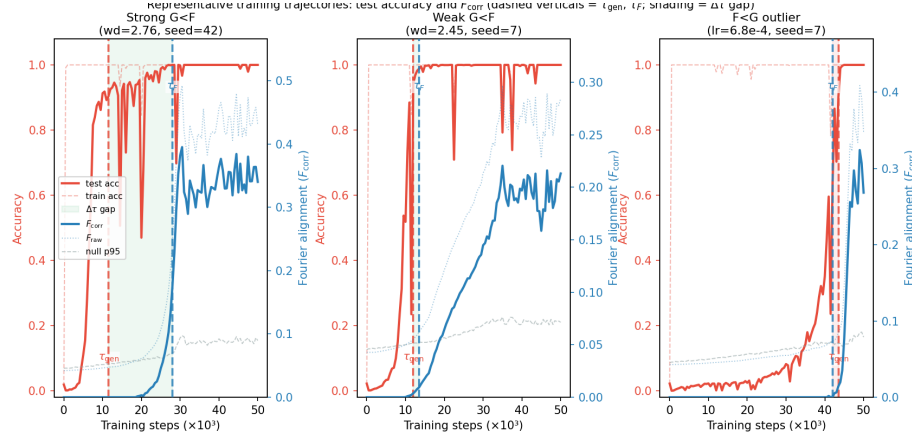


Figure 1: **Example training trajectories for three representative runs.** Red curves: test accuracy (left axis). Blue curves: null-corrected Fourier alignment F_{corr} (right axis, dotted = raw F_{raw} , grey dashed = null p95 baseline). Dashed verticals mark τ_{gen} (red) and τ_F (blue). *Left and centre:* $G < F$ runs where generalization fires first and the green-shaded $\Delta\tau$ gap follows; F_{corr} rises only after test accuracy has already reached 1. *Right:* the original Stage 3 $F < G$ outlier ($\text{lr} = 6.8 \times 10^{-4}$, seed 7), stable under detector hyperparameters but not reproduced in the targeted cross-hardware re-run of Section 5, where F_{corr} barely exceeds the null baseline (0.015) one log-interval before an unusually late $\tau_{\text{gen}} = 43,500$.

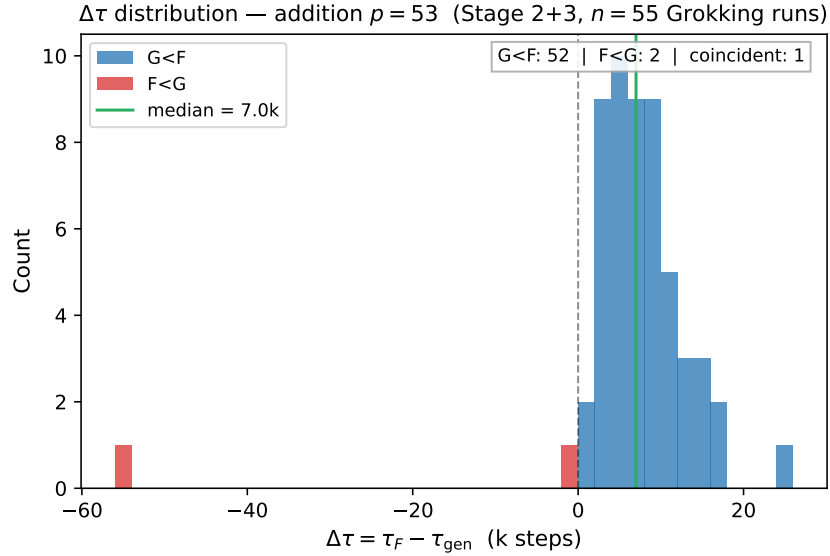


Figure 2: Distribution of $\Delta\tau = \tau_F - \tau_{\text{gen}}$ across all 55 Grokking runs (Stage 2 blue, Stage 3 red). Positive values indicate $G < F$ (generalization first). Median $\Delta\tau \approx 7,000$ steps; 52/55 runs (94.5%) are positive. The two negative values are discussed in Section 3.1.

$G < F$ and 1/30 coincident ($\Delta\tau=0$), *no $F < G$* ; **Stage 3:** 25/30 Grokking (5 Memorization at high lr), **23/25 $G < F$** with two phase-boundary $F < G$ outliers (low- lr : $\text{lr} = 6.8 \times 10^{-4}$, seed 7, $\Delta\tau = -1,500$, not reproduced in the targeted slow-grokking re-run of Section 5; high- lr : $\text{lr} = 4.32 \times 10^{-3}$, seed 42, $\Delta\tau = -54,500$, remains an atypical boundary case). Combined: **52/55 (94.5%, CI 85.1%–98.1%) $G < F$** , median $\Delta\tau \approx 7,000$ steps (Figure 2). Per-cell tables and per-stage scatter plots are in Appendix M.

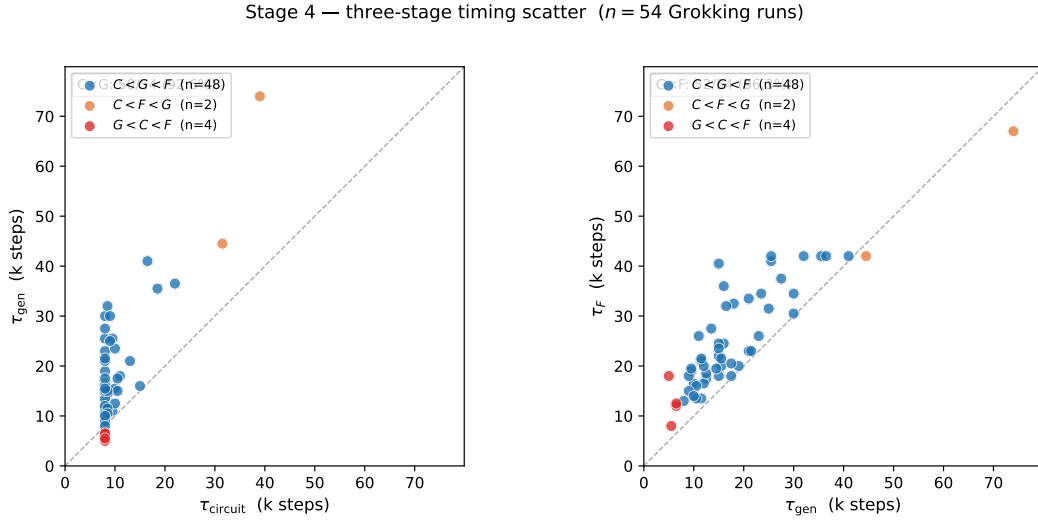


Figure 3: **Three-stage timing scatter (Stage 4).** *Left:* τ_{circuit} (circuit) vs τ_{gen} (generalization). Points above the diagonal ($y > x$) confirm $C < G$ ordering. Blue: $C < G < F$ (dominant); orange: $C < F < G$; pink: $G < C < F$. *Right:* τ_{gen} vs τ_F (embedding geometry). Points above diagonal confirm $G < F$. $C < G$ holds marginally in 50/54 runs (92.6%); the stricter joint $C < G < F$ pattern in Table 1 requires both $C < G$ and $G < F$ in the same run.

Table 1: Stage 4: three-stage timing across Stage 2 ($\text{lr} = 1.6 \times 10^{-3}$) and Stage 3 ($\text{wd} = 2.5$) Grokking runs. Medians in thousands of steps.

Subset	$C < G < F$	$C < G < F$ rate	med. $\tau_{\text{gen}} - \tau_{\text{circuit}}$ (k)	med. $\tau_F - \tau_{\text{gen}}$ (k)
Stage 2 (lr fixed)	30/30	100%	6.3	6.0
Stage 3 (wd fixed)	18/24	75.0%	6.3	6.0
All	48/54	88.9%	6.3	6.0

3.2 STAGE 4: CIRCUIT ONSET τ_{circuit} AND THREE-STAGE ORDERING

We retrained all Stage 2+3 cells with the additional F_L metric, yielding τ_{circuit} for each run. τ_{circuit} was detected in all 54 Grokking runs (100%).

$C \leq G$ (circuit at or before generalization). In 49/54 Grokking runs (90.7%), $\tau_{\text{circuit}} < \tau_{\text{gen}}$ strictly; in one additional run $\tau_{\text{circuit}} = \tau_{\text{gen}}$ (both detected at the same 500-step checkpoint), giving 50/54 (92.6%, CI 82.4%–97.1%) under the $C \leq G$ convention used throughout this section. The median gap $\tau_{\text{gen}} - \tau_{\text{circuit}} \approx 6,250$ steps (range $-3,000$ to $35,000$ steps). The 4 exceptions with $\tau_{\text{circuit}} > \tau_{\text{gen}}$ (i.e. $G < C < F$) occur at the highest learning rates ($\text{lr} \geq 2.3 \times 10^{-3}$, $\text{wd} = 2.5$), where generalization is very fast ($\tau_{\text{gen}} \leq 6,500$ steps) and the logit Fourier signal rises too gradually for BIC to detect a changepoint before τ_{gen} .

Proxy-level three-stage ordering $C < G < F$. The dominant pattern is $C < G < F$ in 48/54 runs (88.9%, CI 77.8%–94.8%). The remaining runs split into $C < F < G$ (2 runs, low-lr boundary) and $G < C < F$ (4 runs, high-lr boundary). The three intervals are nearly equal in magnitude: median $\tau_{\text{gen}} - \tau_{\text{circuit}} \approx 6,250$ steps, median $\tau_F - \tau_{\text{gen}} \approx 6,000$ steps, median $\tau_F - \tau_{\text{circuit}} \approx 13,000$ steps.

3.3 AGGREGATE $G < F$ RATE ACROSS STAGES

Across the main experimental stages, $G < F$ remains the dominant ordering beyond the initial 1D sweeps. Table 2 aggregates the $G < F$ rate across the two tasks ($a+b$ and $a-b$), the two primes ($p \in \{53, 97\}$), and the dense 2D grid (Stage 6), reporting Wilson 95% confidence intervals for each.

The “All (main stages)” row gives the headline 89.8% (212/236); adding the targeted slow-grokking sweep of Section 5 brings the post-hoc aggregate to 236/260=90.8%.

Table 2: $G < F$ rates across all experimental stages. n_G : Grokking runs; $G < F$: ordering rate and median lag. 95% CIs are Wilson intervals for a binomial proportion. The “All (main stages)” row excludes the targeted slow-grokking sweep of Section 5 (which adds 24/24 $G < F$ cells and brings the aggregate to 236/260=90.8%).

Stage	Setting	n_G	$G < F$ rate	95% CI	med. $\Delta\tau$ (k)
2+3	add mod 53, 1D sweeps	55	94.5%	85.1–98.1%	7.0
5A	mul mod 53	30	100%	88.6–100%	8.8
5B	add mod 97	21	100%	84.5–100%	28.0
6	add mod 53, 2D grid	130	83.8%	76.6–89.2%	6.0
All (main stages)		236	89.8%	85.3–93.1%	—
§5	slow-grokking sweep (post-hoc)	24	100%	86.2–100%	11.75
All (incl. slow-grokking sweep)		260	90.8%	86.6–93.7%	—

3.4 CAUSAL FOURIER-SUBSPACE INTERVENTION (E7)

E7 — Causal Fourier-subspace intervention. The $C < G$ ordering reported in Section 3.2 relies on a logit-level Fourier proxy, and the E3 cross-hardware replication in Appendix J.5 shows the $C < G$ rate is detector-dependent. To test whether the dominant low-order Fourier subspace identified by the proxy is causally load-bearing rather than a correlated readout, we probe twelve cells: five original cells spanning the phase diagram (strong $C < G < F$, $G < C < F$ boundary, $C < F < G$ slow-grokking, canonical, coincident) plus seven expansion cells varying seed ($\times 2$), weight decay ($\times 2$), learning rate ($\times 1$), task (multiplication, $\times 1$), and prime ($p=97$, $\times 1$). Each cell is saved at three to four regime-boundary checkpoints (pre-all, between events, post-all). At each checkpoint we apply four interventions on the trained logits: **(I1)** keep only the top-3 Fourier harmonics of the output-class index, **(I2)** remove them, **(I3)** remove a matched-energy random subspace as control, and **(I4)** apply the same Fourier removal to the token embedding rows (anonymised supplementary code).

Three findings emerge (Figure 4 shows a five-cell subset). First, on the eight addition-mod-53 cells that fully grok, the maximum I2 drop across between-event checkpoints reaches 0.40–0.72 in absolute test accuracy (specificity 12–214 $\times$ over matched-energy random controls), demonstrating that the top-3 Fourier subspace is causally load-bearing. The effect extends to $p=97$: top-3 alone is insufficient (I2 drop 0.004) because the larger prime spreads load across more harmonics, but at $K=5$ the drop reaches 0.38 and at $K=10$ it reaches 1.00, all with random-control drops of exactly 0.000. The multiplication cell (mul_GF) is a clean negative: specificity $\approx 1\times$ at every K , indicating that discrete-log reindexing changes which directions carry load; we exclude it from headline statistics. Second, after τ_F the I2 drop on logits collapses while I4 on embeddings stays large, indicating redundant logit-decoding paths develop post- τ_F while the embedding remains Fourier-dependent, consistent with our lagging-indicator interpretation. Third, the $G < C < F$ cell ($G_CF_boundary$, $\tau_{gen}=6,500 < \tau_{circuit}=8,000$) shows Fourier load-bearing already at step 7,000 (drop 0.69, specificity 112 $\times$), reframing the four $G < C < F$ exceptions in Section 3.2 as detector-resolution artefacts.

Why $K=3$? A top- K ablation spectrum. A reasonable concern is whether $K=3$ is privileged or whether removing any low-rank subspace would damage accuracy similarly. We rerun the intervention for $K \in \{1, 3, 5, 10\}$ on all twelve cells, using the same matched-energy random-subspace control at each K . Table 3 reports medians over the fifteen between-events checkpoints with base accuracy > 0.5 (excluding mul_GF). The causal effect grows smoothly with K from $K=1$ (insufficient coverage; I2 drop 0.080) through $K=3$ (drop 0.551) to $K=5$ (drop 0.812), and saturates by $K=10$ (drop 0.974). At every K the matched-energy random control remains within $|\Delta acc| \leq 0.007$ of uninervened accuracy, so the specificity of top- K Fourier removal over a random subspace *rises* from 10 $\times$ ($K=1$) to $\geq 360\times$ ($K=5$). The shape of the curve confirms that the causal effect is localized to low-order logit-Fourier structure rather than to arbitrary low-rank damage.

Table 3: Top- K ablation spectrum on the twelve causal-probe cells (excluding `mul_GF`), restricted to between-events checkpoints with base accuracy > 0.5 ($n=15$). I2 removes the top- K Fourier harmonics; I3 removes a matched-energy random subspace. Specificity is I2 drop / I3 drop.

K	median I2 drop	median I3 ctrl drop	median specificity
1	+0.080	+0.007	$10.4\times$
3	+0.551	+0.005	$37.5\times$
5	+0.812	+0.001	$362\times$
10	+0.974	-0.004	(saturated) [†]

[†]At $K=10$ the random control drop is slightly negative on the median, making the specificity ratio undefined; I2 drop has saturated near 1.0 regardless.

Are the top- K harmonics special, or would any K Fourier directions do? A frequency-selectivity control. The top- K subspace removal still leaves open a second attack: maybe the load-bearing structure is just “any K Fourier directions”, not specifically the high-power ones. We test this directly by comparing, at the same checkpoints and at $K=3$, three alternative ablations that stay *inside* the Fourier basis but pick different frequencies: **top-3** (the three harmonics with the highest centered-logit power, the canonical I2), **alt-3** (three harmonics drawn uniformly from rank ≥ 10 , averaged over 5 random draws), and **bot-3** (the three lowest-power harmonics). For comparison we also report the matched-energy non-Fourier random-subspace control (I3) used throughout this section. Table 4 summarises the result over the fifteen between-events checkpoints (excluding `mul_GF`). Top-3 removal drops accuracy by a median of 0.55; alt-3 and bot-3 removals drop accuracy by -0.003 and -0.002 on the median, statistically indistinguishable from the matched-energy non-Fourier random control (+0.008). The load-bearing logit-Fourier structure is therefore not “any K Fourier directions” but specifically the small set of high-power answer-class harmonics.

Table 4: Frequency-selectivity control at $K=3$ on twelve causal-probe cells (excluding `mul_GF`), restricted to between-events checkpoints with base accuracy > 0.5 ($n=15$). Every variant removes a 3-harmonic Fourier subspace from the centered logits; they differ only in *which* three harmonics are chosen.

Removed subspace ($K=3$)	median accuracy drop
top-3 Fourier (by power)	+0.551
alt-3 Fourier (rank ≥ 10 , mean of 5 random draws)	-0.003
bot-3 Fourier (lowest-power)	-0.002
matched-energy non-Fourier random subspace (I3)	+0.008

4 DISCUSSION

Our results distinguish *mechanism formation* from *embedding consolidation*: the logit-Fourier proxy F_L rises before τ_{gen} (the C<G half of the timing picture, complementing Nanda et al. (2023)), while the embedding-side F_{corr} continues to consolidate for thousands of steps *after* τ_{gen} ($\sim 6\text{k}$ -step gaps between successive onsets). The familiar Fourier ring in PCA visualizations is therefore stage (3), not stage (1).

Limits of the proxy: dissociation is between subspaces, not full mechanisms. We use τ_{circuit} as a logit-side Fourier proxy rather than as a claim that the complete circuit has been reverse-engineered at that step. The full modular-addition algorithm is implemented by a distributed transformer computation involving attention, MLP, and potentially multiple harmonics. Our claim is therefore narrower than “the algorithm forms before generalization”: the low-order answer-class *logit-Fourier subspace* identified by this proxy is already functionally necessary before the *embedding-Fourier ring* consolidates. The dissociation we establish is between an early load-bearing logit-Fourier subspace and a later visible embedding-Fourier geometry, not between a fully specified circuit and an irrelevant embedding matrix. Consistently, under coarser loss-based proxies the C<G rate falls to 63.6%/23.6% (Section J.5, E3), so the proxy-level C<G<F headline is detector-dependent rather than universal. F_only runs further show that embedding alignment is *insufficient* for generalization; E7 itself is

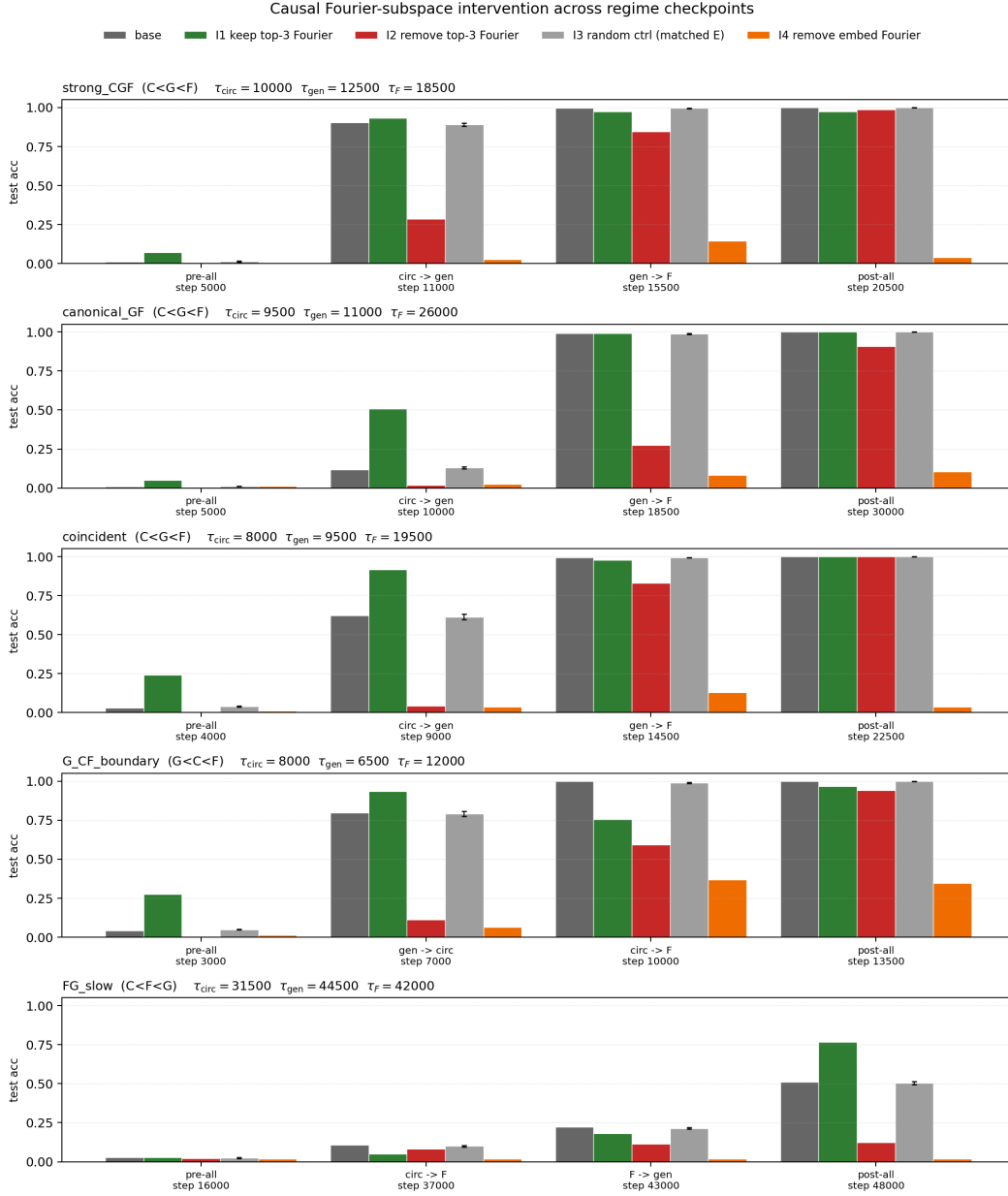


Figure 4: **Causal Fourier-subspace intervention (five-cell subset of the twelve probed cells)**. Each row is one cell; the four bar groups are regime checkpoints (*pre-all*, between-event, *post-all*); each group reports test accuracy under five conditions: unintervened base; I1 keep only the top-3 Fourier harmonics; I2 remove them; I3 a matched-energy random-subspace control; I4 remove top-3 Fourier from the token embedding. The large I2 drops with negligible I3 effect in the between-event regime (specificity 12–214× across all eight fully-grokged addition-mod-53 cells) establish the dominant Fourier subspace as causally load-bearing. The post- τ_F I2 collapse alongside sustained I4 drop indicates redundant logit-decoding paths develop after τ_F while the embedding remains Fourier-dependent.

a 12-cell causal-localization probe (spanning seeds, weight decay, learning rate, task, and prime) whose large- N counterpart is the 236/260 timing aggregate. Connections to rank/flatness/complexity timing, neural-collapse-style consolidation, and superposition are deferred to Appendix J.

Practical implication. Embedding-based progress measures (PCA ring inspection, dictionary-learning probes) may systematically underestimate how far circuit formation has progressed: the logit-Fourier subspace is already load-bearing at checkpoints where the embedding ring has not yet consolidated. A clean Fourier ring is a *lagging* indicator of algorithmic capability, not a concurrent one.

Architecture scope. We do not claim architecture-universality: all main experiments use the same 2-layer transformer ($d_{\text{model}}=256$, 4 heads, $d_{\text{ff}}=1024$). A small post-hoc depth pilot at the canonical operating point reinforces this boundary rather than removing it: with all other hyperparameters fixed, the 1-layer variant gives $F < G$ in all three seeds, and the 3-layer variant gives $F < G$ in two of three seeds (Appendix F, E2). One plausible account is that deeper networks provide alternative pathways that let logit-Fourier structure propagate into embeddings before generalization, collapsing the timing gap. Because this pilot covers only one operating point and two depths, we treat it as a scope warning; a systematic depth $\times$ width transfer study remains the primary future work.

5 FALSIFYING THE STRONG-FORM SPEED-DEPENDENT ORDERING HYPOTHESIS

The Stage 3 $F < G$ outliers concentrate at large τ_{gen} , suggesting a strong-form hypothesis: *slow grokking systematically reverses the ordering to $F < G$ above $\tau_{\text{gen}} \sim 35$ k*. We falsify this with a targeted sweep at $\text{lr} \in \{5.0, \dots, 8.0\} \times 10^{-4}$, $\text{wd} = 2.5$, seeds $\{42, 7, 2025\}$, 100–120 k steps (24 cells). **All 24 grok and all 24 are $G < F$** with $\Delta\tau \in [+2.5, +19.5]$ k (median +11.75 k); the slowest cell ($\tau_{\text{gen}} = 54.5$ k) still has $\Delta\tau = +3$ k. The tercile breakdown by τ_{gen} shows 0/9, 0/7, 0/8 $F < G$ across low/middle/high terciles, falsifying the predicted transition (full scatter: Appendix K, Figure 13). The original Stage 3 outlier at $(\text{lr}, \text{seed}) = (6.8 \times 10^{-4}, 7)$ re-trained on different hardware shifts from $\Delta\tau = -1.5$ k to +9 k (label flip), consistent with seed/hardware variance rather than a systematic regime. Adding the 24 cells brings the aggregate to 236/260=90.8% $G < F$.

6 CONCLUSION

We have separated two events that are easy to conflate in grokking: the emergence of a load-bearing Fourier computation in output-logit space, and the later consolidation of the visible Fourier ring in token-embedding space. Timing and causal evidence converge: the logit-Fourier proxy precedes generalization, the embedding-Fourier onset follows it, and the early logit-Fourier subspace is already causally necessary before the ring has consolidated. In the controlled 2-layer modular-arithmetic setting studied here, the familiar PCA ring is best interpreted as a post-generalization consolidation signal rather than as the structure through which the algorithm first becomes load-bearing.

Ethics Statement. This paper studies controlled algorithmic tasks in small transformers and does not involve human subjects, private data, deployed systems, or new datasets containing sensitive information. We do not identify direct societal harms beyond general dual-use considerations associated with machine learning research.

Reproducibility Statement. Hyperparameters, seeds, detector settings, and hardware/software versions are given in Appendices B–E; an anonymized supplementary package provides per-run CSVs, configs, plotting scripts, and code reproducing every main figure and table. An LLM usage statement is in Appendix A.

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A LLM USAGE STATEMENT

The authors used large language model tools for coding assistance, debugging, language polishing, and discussion of possible alternative explanations. The research questions, experimental design decisions, implementation choices, result validation, interpretation, and final claims were made by the authors. All references, numerical results, and claims were checked by the authors against the code, logs, and cited sources.

B EXPERIMENTAL DETAILS

B.1 BIC CHANGEPOINT ESTIMATOR

We apply an exponential moving average ($\alpha=0.15$) to the F_{corr} trajectory, then operate on $\log(1+t)$. We search all candidate breakpoints in the interior two-thirds of the sequence, fitting OLS regressions before and after each candidate. The 4-parameter breakpoint model is accepted if $\text{BIC}_{\text{break}} < \text{BIC}_{\text{no-break}}$, where $\text{BIC}_k = n \log(\text{RSS}/n) + k \log(n)$. A persistence check requires the post-break slope to exceed 1% of the total value range.

B.2 NULL PERMUTATION ESTIMATION

Each run maintains a fixed NumPy RNG seeded as $\text{run_seed} \oplus 0 \times \text{DEAD}$, independent of the model RNG. We shuffle the p token indices 100 times at every 500-step log point (the canonical refresh cadence; see Section 2) to estimate $F_{\text{null},p95}$. Sensitivity-analysis configurations using 2,500- and 5,000-step refresh intervals are reported in Section J.4.

B.3 HYPERPARAMETER VALUES

Stage 2. $\text{wd} \in \{1.20, 1.35, 1.52, 1.71, 1.93, 2.18, 2.45, 2.76, 3.11, 3.50\}$; $\text{lr} = 1.6 \times 10^{-3}$; seeds = $\{42, 7, 2025\}$.

Stage 3. $\text{lr} \in \{5.00, 6.80, 9.26, 12.6, 17.1, 23.3, 31.7, 43.2, 58.8, 80.0\} \times 10^{-4}$; $\text{wd} = 2.5$; seeds = $\{42, 7, 2025\}$.

C COMPLETE TIMING RESULTS

Figure 5 shows the full scatter of $(\tau_{\text{gen}}, \tau_F)$ for all 55 Grokking runs. Points above the diagonal ($\tau_F > \tau_{\text{gen}}$) correspond to $G < F$; the two red star (★) points below are the $F < G$ cases discussed in Section 3.1. The scatter confirms that the lag $\Delta\tau = \tau_F - \tau_{\text{gen}}$ is distributed throughout the range 1,000–25,000 steps with no obvious bimodal structure.

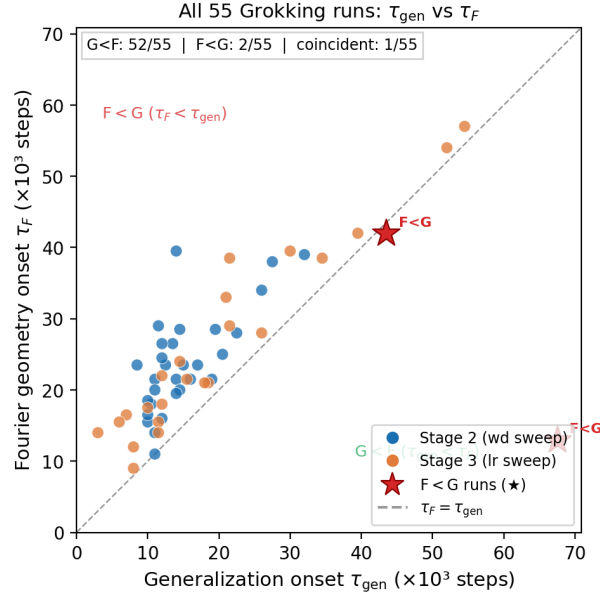


Figure 5: Scatter plot of τ_{gen} vs τ_F for all 55 Grokking runs (blue = Stage 2 wd sweep; orange = Stage 3 lr sweep). All points above the dashed diagonal are $G < F$. The two red star (★) points below the diagonal are the $F < G$ outliers discussed in Section 3.1.

D PER-CELL DETECTOR SENSITIVITY

Figure 6 extends the aggregate robustness heatmap (Figure 10) to show $P(G < F)$ per cell, allowing inspection of which configurations drive the $\text{RS} < 1$ cases.

Figure 7 provides a direct comparison of the $F < G$ outlier trajectory against the strong $G < F$ reference cell.

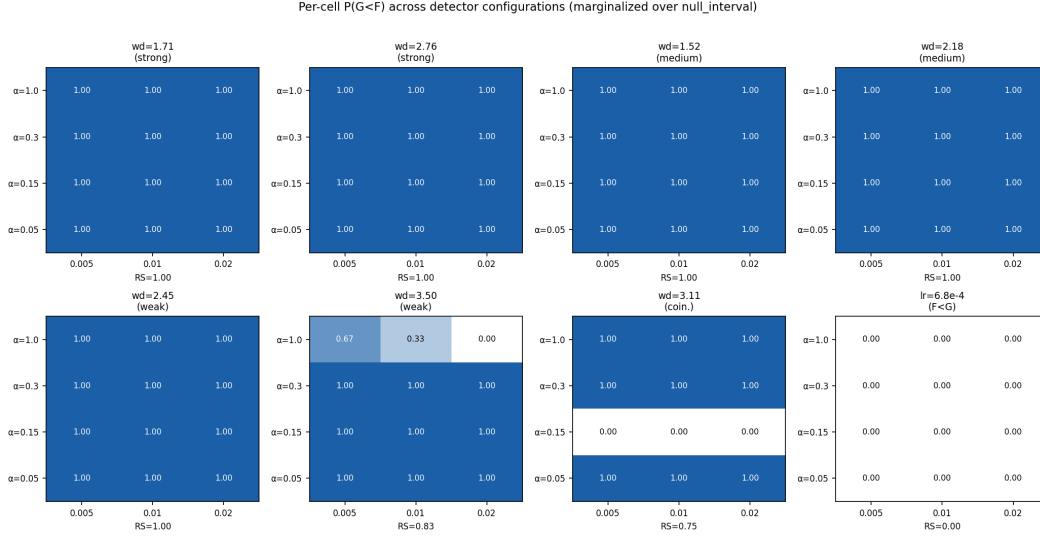


Figure 6: Per-cell $P(G<F)$ across all 36 detector configurations, marginalized over null-update interval. Strong and medium $G<F$ cells (first four panels) are uniformly blue ($RS=1.00$). Weak $G<F$ cells degrade to *coincident* (not $F<G$) under the strictest slope threshold. The $F<G$ outlier (bottom right) is uniformly red across all 36 configurations, confirming that its ordering is stable with respect to detector parameters.

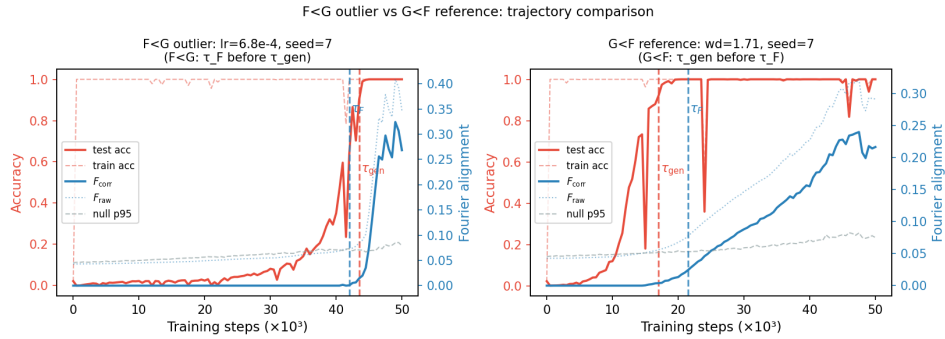


Figure 7: Trajectory comparison: $F<G$ outlier ($lr=6.8 \times 10^{-4}$, seed 7; left) vs. a strong $G<F$ reference ($wd=1.71$, seed 7; right). In the outlier, F_{raw} (dotted blue) hovers near the null baseline (grey dashed) throughout training and F_{corr} only barely exceeds zero near the very late $\tau_{gen} = 43,500$. In the reference cell, F_{corr} shows a clear, sustained rise well after the sharp test-accuracy jump.

E REPRODUCIBILITY

The majority of experiments reported in this paper were run on a local NVIDIA RTX 4080 Laptop (12 GB VRAM); a smaller subset of longer-running sweeps was executed on rented cloud GPU instances (NVIDIA RTX PRO 6000 and NVIDIA A800) accessed remotely via SSH. All runs used the same PyTorch training loop (Python 3.11.9, PyTorch 2.5.1+cu121, NumPy 2.4.3, CUDA 12.1, cuDNN 9.1.0) with identical model and optimiser code, and `torch.backends.cudnn.deterministic=True` is set in the seeding routine. **Bit-exact reproducibility holds within a single GPU/CUDA configuration but not across them.** A direct comparison of two independent sweeps on the same nominal seeds (Stage 4 on a local RTX 4080 vs E3 on a rented A800), documented in Section J.5 (E3), shows per-cell τ_{gen} shifting by a median of 2,000 steps and the three-way ordering label reclassifying on 8/60 cells. Single-run point estimates of $C < G$ and $C < G < F$ rates therefore carry an empirically observed ~ 9 percentage points of cross-hardware run-to-run variance. Each modular-arithmetic training run terminates in 20–45 minutes at 50,000 steps and 30–75 minutes at 80,000 steps on the local 4080; cloud runs are correspondingly faster on the larger data-centre GPUs. The full set of experiments reported here (Stages 2–6 plus the subtraction pilots) totals approximately 280 runs and ~ 170 GPU-hours (aggregated across hardware).

Random seeds. Each cell is run with seeds $\{42, 7, 2025\}$. Seeds are applied uniformly to PyTorch, NumPy, and Python’s `random` at the start of each run. The permutation-null estimator uses an independent NumPy generator seeded with `run_seed  $\oplus$  0xDEAD` so that null resampling is reproducible but decoupled from model initialisation.

Hyperparameter table. Full hyperparameter values for each stage are listed in Appendix B. The canonical detector configuration used for all headline numbers is: null-update interval = 500 steps, EMA smoothing $\alpha = 0.15$, BIC slope threshold = 0.01 of the total value range.

Release plan. An anonymized supplementary package for review includes the training scripts, per-run metric trajectories (F_{corr} , F_L , test accuracy at every 500-step checkpoint, stored as compressed `.npz` files), changepoint scripts, plotting scripts that reproduce every figure and table in this paper, `requirements.txt`, and configuration files for every axis of variation reported. A de-anonymized public repository will be released after acceptance.

F FUTURE WORK

We list two natural next steps that extend the empirical scope of this paper. Neither is required to support the claims in the main body, and both are deferred to future work.

E1. Full subtraction sweep. Replicate Stages 2 and 3 with operation $(a - b) \bmod p$ for $p \in \{53, 97\}$. Expected cost: ~ 60 runs, ~ 45 GPU-hours. The goal is to test whether the $G < F$ rate is better explained by generalization timing than by the task label.

E2. Architecture-depth scan. A small post-hoc pilot already indicates that depth transfer is not automatic: at the canonical operating point, $n_{\text{layers}} = 1$ gives $F < G$ in 3/3 seeds and $n_{\text{layers}} = 3$ gives $F < G$ in 2/3 seeds, while the main paper’s 2-layer setting gives $G < F$ at the same hyperparameter region. This pilot is too narrow to support a depth law, but it is strong enough to make architecture universality an explicit non-claim. A full scan should sweep depth and width jointly across multiple Grokking cells, rather than only the single canonical operating point used in `results/arch_depth/results.csv`.

G BACKGROUND

G.1 GROKKING AND THE PHASE DIAGRAM

Power et al. (2022) documented grokking in small transformers on binary operations including $(a + b) \bmod p$. Liu et al. (2022) mapped training behavior across a (decoder learning rate, weight decay) grid, identifying four phases: *Grokking* (delayed generalization), *Memorization* (no generalization),

Comprehension (immediate generalization), and *Confusion* (no learning). Our experiments follow their architecture and phase conventions.

G.2 FOURIER CIRCUITS AND EMBEDDING GEOMETRY

Nanda et al. (2023) reverse-engineered the algorithm underlying grokked transformers: the network computes modular sums via trigonometric identities over task-relevant Fourier harmonics. They introduced mechanism-level progress measures and found that circuit formation, the process by which the model constructs the Fourier computation, *precedes* the test-accuracy jump, followed by a “cleanup” phase in which memorization components are removed. Their analysis focused on the computational mechanism (logit-level Fourier structure and progress measures); we study *both* that mechanism and a complementary quantity: the geometric organization of the *token embeddings themselves*. We introduce a logit Fourier alignment score F_L that measures circuit-level structure (how much logit variance is explained by Fourier harmonics of the answer class), and compare its onset τ_{circuit} with embedding-geometry onset τ_F in the same runs. Progress measures / F_L track how much of the Fourier *computation* is already in place; F_{corr} tracks when the token *embedding table* has converged to its final geometry.

H RELATED WORK

Grokking and delayed generalization. Power et al. (2022) introduced grokking on modular arithmetic. Liu et al. (2022) constructed the (lr, wd) phase diagram we use as our experimental scaffold, identifying the four training phases we adopt. Liu et al. (2023) showed that grokking extends well beyond algorithmic tasks and is governed by the ratio of initial weight norm to the norm of the generalising solution; Humayun et al. (2024) subsequently argued that grokking-like delayed-generalization dynamics are pervasive even in standard classification networks, and Miller et al. (2024) document the same phenomenon outside neural networks entirely, suggesting the dynamics studied here are an instance of a broader class. Thilak et al. (2022) linked grokking to cyclic instabilities of adaptive optimisers, terming this the “slingshot mechanism.” On the theoretical side, Rubin et al. (2024) model grokking as a first-order phase transition, Lyu et al. (2024) prove that a dichotomy of early- and late-phase implicit biases can provably induce grokking in homogeneous networks, and Mohamadi et al. (2024) give a margin-based theoretical analysis specific to modular addition, directly motivating our use of modular addition as the primary testbed. Kumar et al. (2024) reframe grokking as the transition from “lazy” to “rich” (feature-learning) training dynamics, which provides a candidate mechanism for the long delay between τ_{gen} and τ_F that we measure. Doshi et al. (2024) disentangle generalization from memorization on corrupted modular-arithmetic data, finding that the two events follow distinct trajectories controlled by different hyperparameters; this complements our finding that the logit-level Fourier proxy for circuit formation, generalization, and embedding-geometry consolidation can be temporally pried apart within a single uncorrupted training run. These works characterise *when* or *whether* generalization occurs; our work characterises its timing *relative* to embedding geometry and a logit-level Fourier alignment proxy for circuit formation in the same runs.

Mechanistic interpretability of grokked models. The circuit-level view of neural networks (Olah et al., 2020; Elhage et al., 2021) treats trained models as collections of identifiable computational subgraphs; modular arithmetic has become a flagship setting because the circuit can be fully reverse-engineered. Nanda et al. (2023) reverse-engineered the Fourier circuit and showed via progress measures that circuit formation precedes the generalization jump. Our F_L metric is analogous to their restricted logit loss but computed for all (a, b) pairs simultaneously, providing a single continuous signal for BIC changepoint estimation of τ_{circuit} . Varma et al. (2023) argue that grokking reflects the delayed ascent of an efficient generalising circuit over a more immediately learnable memorising solution; Merrill et al. (2023) make this competition concrete by characterising it as a contest between sparse and dense subnetworks, in which the sparse generalising circuit eventually wins out. This sparse-vs-dense framing predicts a temporal gap between *circuit assembly* and *circuit dominance*, consistent with the $C < G$ separation we measure. Chughtai et al. (2023) study how networks learn group operations more broadly, finding that the learned circuits depend on the group structure in a predictable way. Zhong et al. (2023) show that modular addition admits two distinct algorithms (the “Clock” and “Pizza” variants), depending on initialisation and architecture, revealing that the

Fourier circuit is not the unique generalising solution. Gromov (2023) provides analytic weight solutions for two-layer networks on modular arithmetic tasks, establishing complete interpretability of the learned representations. Conmy et al. (2023) automate circuit discovery via activation patching, complementing our temporal approach to identifying when circuits form. Beyond modular arithmetic, the circuit framework has matured to larger models: induction heads in language models (Olsson et al., 2022), the indirect-object-identification circuit in GPT-2 (Wang et al., 2023), and sparse-autoencoder feature dictionaries (Bricken et al., 2023; Huben et al., 2024) are recent examples of mechanistic explanation at scale. Our F_{corr} metric is complementary to all of the above: it tracks when the token *embedding table* reaches its final geometric arrangement, which lags generalization by thousands of steps in most configurations.

Implicit bias and weight decay. Our use of weight decay as the principal axis of variation echoes a long line of work on the implicit biases that accompany gradient descent. Soudry et al. (2018) proved that gradient descent on exponentially-tailed losses converges to the max-margin solution, identifying a direction-selection bias that operates on a much slower timescale than the loss decrease itself. D’Angelo et al. (2024) examine why explicit weight decay remains necessary in modern transformer training despite the use of adaptive optimisers, and Galanti et al. (2022) prove that SGD with weight decay biases learned representations toward low effective rank. The $G < F$ lag we observe (thousands of steps between generalization and the BIC-detected onset of Fourier-aligned embedding geometry) fits naturally with this picture: once the generalising circuit is in place, decoder-side weight decay continues to reshape the embedding rows indirectly through back-propagated gradients (since $\text{wd}_{\text{emb}} = 0$ in our setting), producing a measurable geometric reorganisation that postdates τ_{gen} .

Embedding geometry and representation dynamics. Saxe et al. (2014) showed that learning in deep linear networks proceeds in discrete stages corresponding to singular value decomposition, an early theoretical account of structured representation formation. Pappan et al. (2020) identified “Neural Collapse,” the terminal convergence of within-class activations to a simplex equiangular tight frame, as a generic post-convergence geometric phenomenon in classification networks. Zhu et al. (2021) prove that Neural Collapse is the unique global optimum of an unconstrained-features model with weight decay, and Kothapalli (2023) survey the broader empirical and theoretical scope of the phenomenon. Our τ_F result is analogous to Neural Collapse in spirit but distinct in content: the Fourier ring geometry of token embeddings is also a *post-generalization* geometric consolidation rather than a prerequisite for generalization, but the converged target is the cyclic-group representation rather than a simplex equiangular tight frame. Elhage et al. (2022) show that polysemanticity and superposition are widespread in transformer features; our focus on the token embedding’s one-dimensional ring structure provides a concrete, measurable geometric ground truth against which timing can be assessed. Barak et al. (2022) demonstrate that abrupt generalization can hide gradual progress in loss-scale metrics, motivating our BIC changepoint approach to detect latent structural transitions.

Algorithmic learning in transformers. Modular addition is one of several algorithmic tasks where small transformers have been studied as testbeds for mechanistic interpretation. Lee et al. (2024) examine how data formatting and curriculum choices govern whether small transformers learn integer arithmetic at all; Quirke & Barez (2024) give a circuit-level account of two-digit addition; and Charton (2023) interprets the representations a transformer uses to compute the greatest common divisor. These works confirm that algorithmic tasks share recurring circuit-formation motifs, suggesting our temporal three-stage framework, measured here on $(a + b) \bmod p$ and $(a \cdot b) \bmod p$, is a plausible candidate for application to those settings.

Methods. We use AdamW (Loshchilov & Hutter, 2019) and adopt the transformer architecture of Vaswani et al. (2017) as extended to the circuit framework of Elhage et al. (2021). Our changepoint estimator is based on the Bayesian Information Criterion (Schwarz, 1978); Truong et al. (2020) survey alternative offline changepoint detectors, and the PELT algorithm of Killick et al. (2012) provides a more general optimisation procedure than the single-break search we adopt here. We chose the simpler 1-break formulation because it matches the qualitative shape of F_{corr} (a flat baseline followed by a single rising segment) and admits an interpretable persistence check via the post-break slope. A 2-break sanity check on the same 8 representative cells (Section J.5, “Detector-specification sensitivity”) shows that allowing a second breakpoint does add a late saturation knee that BIC slightly

prefers, but leaves the early breakpoint within median 1,500 steps of the canonical 1-break τ_F ; we view PELT-style multi-break estimation as a refinement that resolves the saturation knee directly rather than as a correction to the timing claim.

Alternative geometric alignment metrics. Several recent works diagnose representation geometry on algorithmic tasks through group-representation theory rather than through a Fourier projection on the answer class. Chughtai et al. (2023) cast modular-arithmetic solutions in terms of irreducible representations of the cyclic group, showing that the network learns specific irreps and that universality across architectures is partly explained at the group-theoretic level. Zhong et al. (2023) identify a complementary “pizza” representation alongside the canonical clock, and show that the choice between the two depends on architectural details of the model. Our F_{corr} and F_L are best read as cheap, single-harmonic proxies for the dominant cyclic irrep on $(a + b) \bmod p$: they trade the generality of full group-representation diagnostics for a scalar timing signal that the BIC pipeline can locate as a single changepoint. The three-stage timing claim in this paper does not rely on F_{corr} being the optimal geometric metric; any monotone geometric alignment score on the cyclic-group representation that crosses a stable threshold should yield a τ_F within measurement resolution of ours, and we expect group-representation diagnostics in the style of Chughtai et al. (2023) to give a compatible but architecture-aware refinement.

Concurrent and adjacent work. Several recent papers study representation-level dynamics in modular-arithmetic grokking and partially overlap with our measurement framework, but each addresses a different question. Khaliq (2026) characterises a post-grokking “scaffold” phase in a 2-layer embedding MLP and argues that the post-collapse embedding harmonic structure causally constrains attractor selection. Our claim is logically compatible but concerns a different temporal question: we do not argue that embedding geometry is irrelevant after grokking; rather, the visible embedding-Fourier ring is not where the load-bearing Fourier computation resides *at the moment of generalization*, while post- τ_F embedding structure may still help determine which grokked attractor the system stabilizes into. Truong Xuan Khanh et al. (2026) report that representation-covariance spectral entropy collapses *before* generalization and couples to the emergence of Fourier-aligned representations; their observable (covariance spectral entropy) differs from our single-harmonic answer-class logit-Fourier subspace, and they do not separate logit-side from embedding-side timing. Xu (2026) use commutator-defect/curvature precursors that precede generalization across learning-rate regimes including a slow regime; their precursor is geometric in weight space and their causal interventions act on gradient flow, not Fourier subspaces. AlquBoj et al. (2025) identify embedding update dynamics and bilinear coupling as grokking mechanisms but do not measure embedding-Fourier alignment timing. Swaroop (2026) analyses ReLU-MLP modular addition and argues that the algorithmic structure is substantially encoded during memorization, an MLP analog of our “logit-Fourier computation forms before generalization” claim. Wang et al. (2026) localise grokking transitions using spectral and Wasserstein diagnostics on representation covariance; their diagnostics target the *generalization transition* as a single event rather than separating logit-side computation from embedding-side geometry, so their measurement is complementary to the three-onset decomposition $(\tau_{\text{circuit}}, \tau_{\text{gen}}, \tau_F)$ we develop here.

Positioning. Table 5 summarises what each of the most closely related works measures. To our knowledge, ours is the first study to measure all three as onset-detected events in the same runs and pair them with a time-resolved logit-Fourier intervention that distinguishes the load-bearing logit-Fourier subspace from the visible embedding ring.

I EXTENDED ROBUSTNESS STUDIES

I.1 STAGE 5A: MODULAR MULTIPLICATION

All 30 runs with $(a \cdot b) \bmod 53$ land in the Grokking phase. **30/30 (100%, CI 88.6%–100%) show G<F ordering** after discrete-log reindexing of token embeddings. Median $\Delta\tau = 8,750$ steps (range 3,500–24,000), comparable to the addition results. The perfect G<F rate extends the finding beyond the addition task, suggesting the ordering is not specific to the $(a + b) \bmod p$ circuit structure.

Table 5: What prior work measures on grokked modular-arithmetic transformers, relative to the three-stage picture. “✓” = directly measured and reported as a timing event; “~” = qualitatively discussed but not measured as an onset time; “.” = not considered. To our knowledge, ours is the first to measure all three in the same runs.

Work	τ_{circuit} (circuit)	τ_{gen} (generalization)	τ_F (embedding geometry)
Power et al. (2022)	.	✓	.
Liu et al. (2022)	.	✓	.
Nanda et al. (2023)	✓	✓	~
Varma et al. (2023)	~	✓	.
Liu et al. (2023)	.	✓	~
Rubin et al. (2024)	.	✓	.
Lyu et al. (2024)	.	✓	.
Doshi et al. (2024)	.	✓	.
This work	✓	✓	✓

Evaluation-domain check. τ_{gen} in the Stage 5A headline is computed from the standard 70/30 random split over the full p^2 input pairs (including pairs with a zero operand), whereas F_{corr} and F_L (and hence τ_F and τ_{circuit}) are restricted to the $\mathbb{F}_p^*$ subgrid (Section 2, “Stage 5A — modular multiplication”). To verify that this evaluation-domain mismatch does not inflate the reported G<F rate, we re-ran the full 30-cell Stage 5A sweep with identical seeds and additionally tracked test accuracy on the $\mathbb{F}_p^*$ -only test subset ($a, b \in \{1, \dots, p-1\}$) at every 500-step checkpoint (src/sweeps/stage5a_fpstar_recompute.py; results in results/stage5a_fpstar/results.csv). The recompute confirms that the matched-domain τ_{gen} is almost identical to the full-split τ_{gen} : $\tau_{\text{gen}}^{\mathbb{F}_p^*} - \tau_{\text{gen}}^{\text{full}} \in \{0, 500\}$ steps for all 30 cells (median 0, max 500, i.e. at most one 500-step measurement bin), the median $\Delta\tau$ is unchanged at 8,750 steps, the minimum $|\Delta\tau^{\mathbb{F}_p^*}|$ is 3,500 steps, and the 30/30 G<F rate is preserved (no cell flips to coincident or F<G). The Stage 5A G<F claim is therefore not an artefact of the evaluation-domain mismatch; the same caveat does not apply to Stage 5B, Stage 2, Stage 3, Stage 4, or Stage 6, all of which use the full-vocabulary additive group $\mathbb{Z}/p$ and have no analogous trivial subgrid.

I.2 STAGE 5B: DIFFERENT PRIME ($p=97$)

Of the 30 runs with $(a+b) \bmod 97$, 21 land in the Grokking phase (9 in Comprehension, concentrated at high wd where fast generalization prevents a Grokking-phase classification). **21/21 Grokking runs (100%, CI 84.5%–100%) show G<F ordering.** The median lag $\Delta\tau = 28,000$ steps is markedly larger than for $p=53$ ($\approx 7,000$ steps), consistent with the larger prime requiring more training steps and a correspondingly longer consolidation period. The G<F rate and direction are unchanged.

I.3 STAGE 6: G<F ACROSS THE FULL 2D PHASE DIAGRAM

The 7×7 grid yields 130 Grokking runs (88.4% of 147 total; 16 Memorization, 1 Comprehension). **109/130 Grokking runs (83.8%, CI 76.6%–89.2%) show G<F ordering** under the canonical $|\Delta\tau| \leq 500$ coincident threshold; 13/130 show F<G, 3/130 are coincident, and 5/130 have G_only (τ_F not detected). Median $\Delta\tau \approx 6,000$ steps.

The G<F rate is not uniform across the grid. By learning rate, G<F peaks at intermediate lr ($\approx 95\%$ for $\text{lr} \in [10^{-3}, 2 \times 10^{-3}]$) and decreases toward both boundaries (58% at $\text{lr} = 5 \times 10^{-4}$, 78% at $\text{lr} = 4.3 \times 10^{-3}$). By weight decay, G<F increases monotonically from 72% at $\text{wd} = 1.43$ to 95% at $\text{wd} \geq 2.93$. F<G cases concentrate at the low-lr and high-lr boundaries (12 of 14 cases under the original detector-only ordering label, 11 of 13 under the canonical $|\Delta\tau| \leq 500$ coincident threshold). This boundary clustering initially motivated the speed-dependent ordering hypothesis tested in Section 5; however, the targeted slow-grokking sweep there falsifies the strong form of that hypothesis (24/24 fresh G<F cells in the predicted F<G margin), and the cross-hardware re-run of the $(\text{lr}, \text{seed}) = (6.8 \times 10^{-4}, 7)$ cell replicates as G<F. We therefore treat these Stage 6 boundary cases as phase-margin irregularities (likely seed/hardware-level fluctuations and detector-resolution

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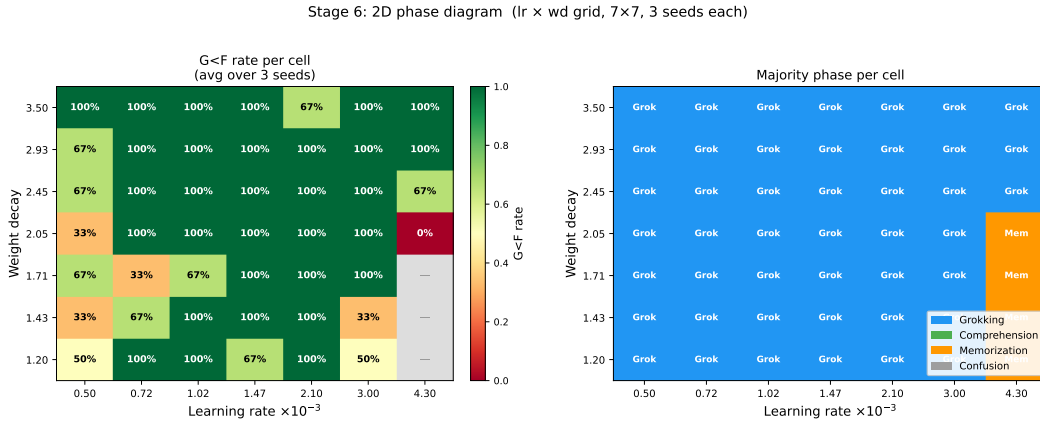


Figure 8: **Stage 6: 2D phase diagram and G<F rate.** *Left:* G<F rate per (lr, wd) cell, averaged over 3 seeds. G<F is strongest in the interior ($\text{lr} \in [10^{-3}, 2 \times 10^{-3}]$, $\text{wd} \geq 2.5$) and weakens toward phase boundaries. *Right:* majority phase per cell. Memorization occupies the high-lr, low-wd corner.

artefacts) rather than evidence for a systematic $F < G$ regime. The interior of the Grokking phase is robustly $G < F$. Figure 8 shows the full spatial structure.

I.4 DECOUPLED WEIGHT-DECAY ABLATION

The canonical sweep applies weight decay only to the decoder parameter group; embedding parameters are by default excluded. To distinguish whether the $G < F$ ordering is driven by decoder weight decay, by embedding weight decay, or by their interaction, and to localise *when* the decay pressure matters, we run two coupled ablations at the canonical operating point $(\text{lr}, \text{wd}^*) = (1.6 \times 10^{-3}, 2.5)$ with $\tau_{\text{gen}}^{\text{ref}} = 17,000$ (source: `src/sweeps/e4_decoupled_wd.py`, results in `results/e4/results.csv`).

2 × 2 ablation — $(\text{wd}_{\text{dec}}, \text{wd}_{\text{emb}}) \in \{0, \text{wd}^*\}^2$. Three seeds per cell, 12 runs total. Table 6 reports the per-cell phase distribution and ordering. Three observations follow, each conditional on the canonical (lr, wd^*) operating point and 3 seeds per cell. **(i) $\text{wd}_{\text{dec}} = 0$ abolishes grokking under this canonical setting:** both cells with zero decoder decay memorise in all 3 seeds (max test accuracy ≤ 0.10). Decoder weight decay therefore appears *necessary* for grokking under this canonical operating point; we read this as a local causal statement, not a universal one. The $(0, \text{wd}^*)$ cell is informative for a separate reason: despite direct weight decay on the embedding parameters, it produces τ_F but no τ_{gen} (F_{only} phase). This provides direct evidence that embedding Fourier alignment is *not sufficient* for generalisation; we deliberately do not claim it is *necessary*, because the headline $G < F$ result of this paper places τ_F *after* τ_{gen} in 212/236 Grokking runs, ruling out the temporal-prerequisite reading of necessity. **(ii) $\text{wd}_{\text{dec}} = \text{wd}^*$ alone reproduces canonical $G < F$:** the $(\text{wd}^*, 0)$ cell groks in all 3 seeds with mean $\Delta\tau = 6,500$ steps, matching the Stage 2 $G < F$ plateau. **(iii) Adding $\text{wd}_{\text{emb}} = \text{wd}^*$ on top of $\text{wd}_{\text{dec}} = \text{wd}^*$ destabilises the canonical dynamics:** 1/3 seeds memorises, and the 2/3 that grok do so with $\tau_F \in \{8,000, 10,000\}$, arriving *before* $\tau_{\text{gen}} \in \{19,500, 16,000\}$ — an $F < G$ ordering, reproducibly. Direct embedding decay therefore accelerates Fourier consolidation enough to precede generalisation, giving a second reproducible recipe for $F < G$ beyond the low-lr regime observed in Stage 3. Taken together, the 2×2 results reframe our central claim: *decoder-only weight decay* is the operating point under which $G < F$ is observed; “weight decay drives grokking” is more accurately stated as “*decoder* weight decay drives grokking, and combining it with embedding decay flips the ordering”.

Decoder-WD shutoff schedule. Starting from the canonical $(\text{wd}_{\text{dec}}, \text{wd}_{\text{emb}}) = (\text{wd}^*, 0)$ configuration, we set $\text{wd}_{\text{dec}} = 0$ at step $s^* \in \{0.5, 1, 2\} \cdot \tau_{\text{gen}}^{\text{ref}}$ (3 seeds each, 9 runs). Results in Table 7 localise the causal window. Shutoff at $0.5\tau_{\text{gen}}^{\text{ref}} = 8,500$, *before* the canonical τ_{gen} , eliminates grokking in all 3 seeds: max test accuracy plateaus at 0.47–0.73 (still a partial progress signal, but never crossing the 0.9 threshold used for τ_{gen}). Shutoff at $\tau_{\text{gen}}^{\text{ref}}$ and at $2\tau_{\text{gen}}^{\text{ref}}$ both preserve canonical grokking, with

Table 6: E4a — 2×2 weight-decay ablation (3 seeds per cell). Phase counts are over the 3 seeds. For Grokking seeds the range of $(\tau_{\text{gen}}, \tau_F, \Delta\tau)$ is reported. Source: `results/e4/results.csv`.

wd_{dec}	wd_{emb}	phase (3 seeds)	τ_{gen}	τ_F	ordering
0	0	Mem $\times 3$	—	—	none
0	wd^*	Mem $\times 3$	—	[8,000, 28,500]	F_only
wd^*	0	Grok $\times 3$	[10,000, 13,500]	[15,000, 23,500]	G<F
wd^*	wd^*	Mem+Grok $\times 2$	[16,000, 19,500]	[8,000, 10,000]	F<G

Table 7: E4b — Decoder-WD shutoff schedule (3 seeds each) at $\tau_{\text{gen}}^{\text{ref}}=17,000$. Source: `results/e4/results.csv`.

shutoff step	phase (3 seeds)	τ_{gen} range	τ_F range	max test acc
8,500 ($0.5\tau_{\text{gen}}^{\text{ref}}$)	Mem $\times 3$	—	[21,500, 24,000]	0.47–0.73
17,000 ($\tau_{\text{gen}}^{\text{ref}}$)	Grok $\times 3$	[10,000, 13,500]	[16,000, 17,500]	1.00
34,000 ($2\tau_{\text{gen}}^{\text{ref}}$)	Grok $\times 3$	[10,000, 13,500]	[16,000, 29,500]	1.00

G<F in 3/3 seeds at each shutoff time. Decoder weight decay therefore appears to act *locally causally* during the pre- τ_{gen} window: removing it within $[0, \tau_{\text{gen}})$ abolishes grokking, while shutting it off after τ_{gen} does not rescind generalisation. Prolonged decay past τ_{gen} delays τ_F (median τ_F at shutoff $2\tau_{\text{gen}}^{\text{ref}}$ is 29,000 vs. 17,000 at $\tau_{\text{gen}}^{\text{ref}}$), consistent with decay continuing to suppress the magnitude signal that drives the BIC changepoint.

Together these two ablations provide *local causal evidence* that decoder weight decay is an optimizer-side condition required for grokking under the tested conditions: within the canonical operating point $(\text{lr}, \text{wd}^*) = (1.6 \times 10^{-3}, 2.5)$ on $(a+b) \bmod 53$ with the 2-layer transformer used throughout and $n=3$ seeds per cell, removing decoder weight decay abolishes grokking, retaining only decoder weight decay (with $\text{wd}_{\text{emb}}=0$) reproduces the canonical G<F regime, and shutting decoder decay off after τ_{gen} does not rescind generalisation. We frame these as local causal observations rather than universal necessity / sufficiency claims: the intervention pattern in deeper transformers, on non-modular tasks, or under different optimisers may shift the causal frontier, and we cannot rule out alternative grokking solutions that operate without decoder weight decay outside our tested grid. Addition of embedding decay on top of decoder decay is not neutral. It flips the ordering to F<G, which restricts the G<F claim to the decoder-only decay regime used throughout Stages 2–6.

I.5 DATA-SUBSAMPLING SPEED CONTROL

E4 provides local causal evidence that decoder weight decay is an optimizer-side condition required for grokking; E5 asks whether the G<F ordering is tied to the absolute value of τ_{gen} (“how late generalisation occurs on the weight-decay clock”) or only to its sign relative to τ_F . Holding the canonical operating point $(\text{lr}, \text{wd}_{\text{dec}}, \text{wd}_{\text{emb}}) = (1.6 \times 10^{-3}, 2.5, 0)$ fixed, we vary the training-set fraction $\rho \in \{0.20, 0.30, 0.40, 0.50\}$ of all p^2 pairs and run 3 seeds per cell (12 runs, 50,000 steps each; source: `src/sweeps/e5_data_subsample.py`, results in `results/e5/results.csv`). Varying ρ changes the data-coverage bottleneck while leaving the architecture, optimizer, learning rate, and weight-decay schedule fixed, providing a partial intervention on τ_{gen} that is less confounded with the lr/wd phase location than tuning the optimiser hyperparameters directly. We acknowledge that varying ρ also alters train/test composition and the memorisation burden on the model, so the intervention is not a clean isolation of the “generalisation clock”.

Table 8 summarises the per- ρ outcome. Three observations follow. (i) At $\rho=0.20$ the model memorises in 3/3 seeds (max test accuracy ≤ 0.02); τ_F is still detected in 2/3 seeds, yielding an F_only phase that replicates the $(\text{wd}_{\text{dec}}=0, \text{wd}_{\text{emb}}=\text{wd}^*)$ cell of Table 6 — Fourier alignment is again confirmed insufficient for generalisation, here bottlenecked by data volume rather than by the decay target. (ii) At $\rho=0.30$ the model enters the canonical grokking window (median $\tau_{\text{gen}}=13,000$, median $\tau_F=17,500$, median $\Delta\tau=5,000$), overlapping the Stage 2 bootstrap interval (median $\Delta\tau=7,750$, 95% CI [6,000, 9,750]; see Table 13). (iii) At $\rho \in \{0.40, 0.50\}$ the model exits the delayed-generalisation

Table 8: E5 — Data-subsampling speed control (3 seeds per ρ , canonical (lr, wd*, 0)). Medians are over seeds with the timescale defined; “–” indicates the timescale is not detected. $P(G<F)$ is the fraction of the 3 seeds with $G<F$ ordering; for $\rho=0.20$ all three seeds are Memorization, so τ_{gen} is undefined and the ratio is not applicable (denoted “N/A” in the table). Source: results/e5/results.csv.

ρ	phase (3 seeds)	med. τ_{gen} (steps)	med. τ_F (steps)	med. $\Delta\tau$ (steps)	$P(G<F)$
0.20	Mem $\times 3$	–	21,750	–	N/A
0.30	Grok $\times 3$	13,000	17,500	5,000	1.00
0.40	Comp $\times 2$, Grok $\times 1$	2,000	20,000	18,000	1.00
0.50	Comp $\times 2$, Grok $\times 1$	1,500	12,000	10,500	1.00

regime: τ_{gen} collapses by roughly an order of magnitude to $\{1,500, 2,000\}$ (Comprehension phase in 4/6 seeds), while τ_F shifts less, and the gap $\Delta\tau$ widens to $\{10,500, 18,000\}$.

The headline result is that τ_{gen} varies by roughly $10\times$ across the four ρ values while the ordering label is $G<F$ in 9/9 runs (3 seeds each at $\rho \in \{0.30, 0.40, 0.50\}$) where τ_{gen} is defined (100%); the $\rho=0.20$ row contributes no Grokking runs and is therefore not part of this denominator. This is the cleanest available dissociation of the $G<F$ ordering from the absolute value of τ_{gen} at the canonical operating point: neither a slower-than-canonical τ_{gen} (as in $\rho=0.30$) nor a faster-than-canonical τ_{gen} (as in $\rho \in \{0.40, 0.50\}$) moves the ordering off $G<F$, under a manipulation that leaves the weight-decay clock untouched.

E5 therefore supports only the *weak* form of the speed-dependent hypothesis of Section 5, namely that $G<F$ is not an artefact of the canonical τ_{gen} value. It does not test the *strong* form, which predicts an $F<G$ transition once τ_{gen} enters the slow-grokking margin above roughly 35,000 steps: all reachable E5 cells have $\tau_{\text{gen}} \leq 13,000$ steps, an order of magnitude below that regime. The strong form requires a targeted slow-grokking test, which we perform in Section 5: that sweep falsifies the predicted transition to a systematic $F<G$ regime (24/24 $G<F$ across $\tau_{\text{gen}} \in [23,000, 54,500]$ steps).

J EXTENDED DISCUSSION

J.1 CONNECTION TO RANK, FLATNESS AND COMPLEXITY TIMING

Our τ_F marks a discrete BIC-detectable event in a single-harmonic projection of the embedding matrix. A growing literature reports that grokking is accompanied by late-phase reductions in solution complexity along several distinct axes: effective rank of weight matrices and Hessian spectra collapsing toward a low-dimensional subspace (Mohamadi et al., 2024), and progress measures based on loss-landscape and feature-statistic transitions (Miller et al., 2024). Our F_{corr} trajectory is consistent with this family of observations because the Fourier subspace alignment we measure is precisely a low-rank reorganisation of the embedding row space onto a 2-dimensional cyclic-group invariant subspace, so a sharp rise in F_{corr} implies an accompanying drop in the effective rank of the embedding restricted to its task-relevant subspace. Two predictions follow. First, in larger transformers solving cyclic-group tasks, the BIC changepoint of F_{corr} should approximately co-locate with the step at which the embedding matrix’s stable rank or participation ratio drops, providing a falsifiable bridge between our single-harmonic onset and rank-based complexity measures. Second, on tasks where the late-phase rank collapse is driven by non-Fourier structure (for example, the Toy Models of Superposition basis-rotation phenomena (Elhage et al., 2022)) we would expect τ_F to lag or fail to detect, while a rank-based progress measure would still register the transition; this provides an experimentally clean separation between cyclic-group-specific and rank-generic notions of “circuit consolidation”.

J.2 WHY THE METRIC MATTERS

We use null-corrected F_{corr} rather than raw F_{raw} because random embeddings already exhibit non-trivial F_{raw} in high dimensions ($p = 53$, $d = 256$): any projection of 53 points in 256 dimensions partially aligns with some harmonic by chance. The null-corrected F_{corr} removes this baseline and marks the step at which embeddings show *significantly above-chance* harmonic structure.

To assess whether null correction changes the conclusions of this paper, we applied the BIC change-point estimator independently to F_{corr} and F_{raw} for all 8 sensitivity cells (Table 9). For cells with $|\Delta\tau| \geq 3,000$ steps, the two metrics agree: 6 of 6 $G < F$ cells remain $G < F$ under raw Fourier, because the sharp post- τ_{gen} rise in F_{raw} dominates the changepoint detection regardless of baseline. The null correction matters most for marginal cases: the coincident cell (wd= 3.11) and the $F < G$ outlier remain unchanged. We conclude that null correction does not alter the main finding for runs with substantial lags, but is conceptually important for calibrating the detection threshold and avoiding false positives in the coincident regime.

Table 9: Ordering under null-corrected vs. raw Fourier for 8 sensitivity cells (canonical detector config). For cells with $|\Delta\tau| \geq 3,000$ both metrics agree; null correction is relevant primarily near the coincident boundary.

Cell	Category	Order (F_{corr})	Order (raw)	Flipped?
wd=1.71, s7	strong $G < F$	$G < F$	$G < F$	No
wd=2.76, s42	strong $G < F$	$G < F$	$G < F$	No
wd=1.52, s42	medium $G < F$	$G < F$	$G < F$	No
wd=2.18, s42	medium $G < F$	$G < F$	$G < F$	No
wd=2.45, s7	weak $G < F$	$G < F$	$G < F$	No
wd=3.50, s7	weak $G < F$	$G < F$	$G < F$	No
wd=3.11, s2025	coincident	coincident	coincident	No
lr=6.8e-4, s7	$F < G$	$F < G$	$F < G$	No
Total		6/8 $G < F$	6/8 $G < F$	0/8

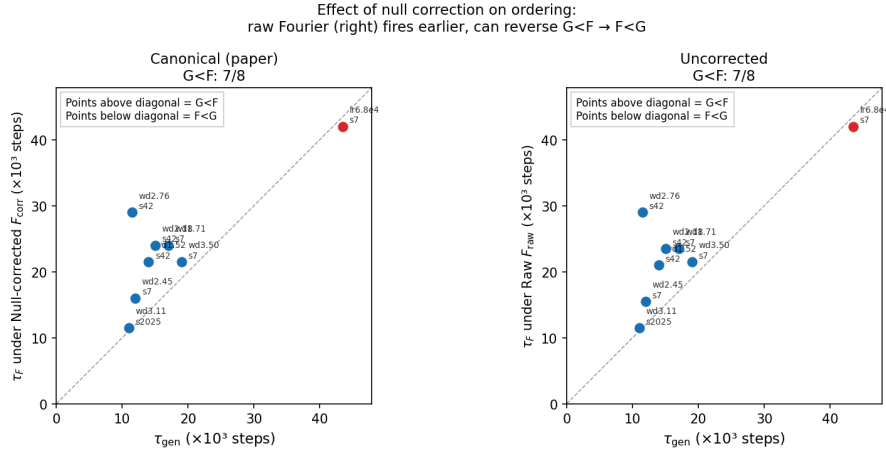


Figure 9: Scatter of τ_F under null-corrected (left) vs. raw (right) Fourier for 8 sensitivity cells. Points above the diagonal indicate $G < F$ ordering; points below indicate $F < G$. Blue points remain $G < F$ under both metrics; the red point ($F < G$ outlier) remains below the diagonal. No ordering flips across the two metrics.

J.3 CONNECTION TO SUPERPOSITION AND POLYSEMANTICITY

The $C < G < F$ ordering carries a tentative implication for representation analysis in larger models. Stage C (τ_{circuit}) marks the BIC-detected onset of single-harmonic Fourier alignment in the logit space, our proxy for circuit formation; Stage F (τ_F) marks the later moment at which the token embedding itself collapses onto the cyclic-group representation that this proxy already detects in the logits. In our 2-layer transformer the two events are separated by 5,000–10,000 steps. The natural reading is that, during the $C < G$ window, the network already implements the correct algorithm but does so on representations that are still polysemantic: the embeddings have not yet specialised onto the cyclic-group basis, so the circuit must operate through a superposition of features rather than through a clean one-feature-per-direction code. This is consistent with the broader picture

of superposition in transformers (Elhage et al., 2022) and with the observation that monosemantic features can be recovered from polysemantic activations only with explicit disentangling (Bricken et al., 2023; Huben et al., 2024). We do not extrapolate this directly to larger models; the present paper measures only a 2-layer transformer on modular arithmetic, and any analogue of τ_F -style geometry consolidation in larger models would have to be established empirically. The narrow consequence we do retain is internal: on this same model, between τ_{circuit} and τ_F the circuit operates on a representation that the embedding-side metric F_{corr} has not yet identified as cyclic-group-aligned, so methods that rely on the embedding’s geometric handle (linear probes, dictionary learning) may underdetect the circuit during that window even though the logit-level Fourier proxy already flags it.

J.4 DETECTOR ROBUSTNESS

A natural concern is whether the $G<F$ finding depends on the specific choice of detector hyperparameters: the null-update interval, the EMA smoothing coefficient, and the BIC slope threshold. The null-update interval axis doubles as the artefact check for the LOCF (piecewise-constant) null schedule introduced in Section 2: a factor-of-10 change in the refresh cadence bounds the maximum τ_F displacement that the LOCF discontinuities could induce. To address this, 8 cells are selected from the Stage 2+3 classification table by a fixed stratification rule: cells stratified across the measured $|\Delta\tau|$ distribution (large, medium, small), plus the coincident cell and the original Stage 3 $F<G$ outlier (detector-stable but not replicated under the cross-hardware re-run of Section 5), and all $3 \times 4 \times 3 = 36$ detector configurations are applied post-hoc to the same saved trajectories. Selecting the subset by a stratification rule applied to the already-classified table, rather than picking individual cells after inspecting their detector behaviour, reduces the risk of cherry-picked robustness.

We define the **Robustness Score** $RS = N_{G<F}/N_{\text{total}}$ for each cell, where $N_{\text{total}} = 36$. Results are shown in Figure 10. The five cells with $|\Delta\tau| \geq 5,000$ steps achieve $RS = 1.00$ (36/36 configurations yield $G<F$). The two cells with $|\Delta\tau| \approx 2,000$ –3,000 steps achieve $RS = 0.83$ and $RS = 0.75$, with the remaining configurations classifying as *coincident* rather than $F<G$. Crucially, **no $G<F$ cell is ever classified as $F<G$** under any configuration. The mean RS across all seven non- $F<G$ cells is **0.94**, confirming that the $G<F$ finding forms a robustness plateau over the detector parameter space.

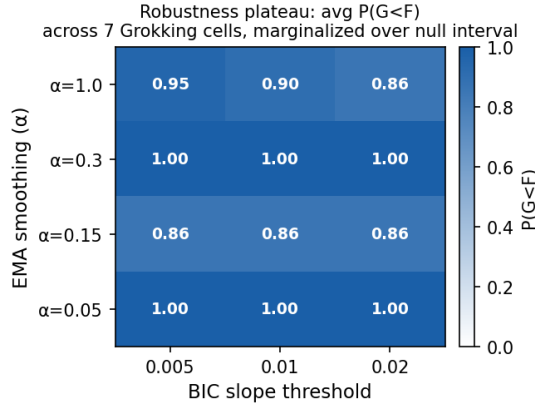


Figure 10: **Robustness plateau.** Each cell shows $P(G<F)$ for a given (BIC slope threshold, EMA α) combination, averaged over null-update intervals $\{500, 2500, 5000\}$ and over all seven Grokking cells. Strong and medium $G<F$ cells achieve 1.00 across all configurations; weaker cells ($|\Delta\tau| \lesssim 3,000$ steps) degrade gracefully to *coincident*, never to $F<G$.

The two $F<G$ cases. Honest treatment requires acknowledging both $F<G$ runs as genuine within the original trajectories and detector pipeline, while characterizing what they reveal. Section 5 additionally shows that the key low- lr $F<G$ case ($lr = 6.8 \times 10^{-4}$, seed 7) does not replicate under an independent cross-hardware re-run, so we treat the two $F<G$ cases as detector-stable but trajectory-level outliers rather than as evidence for a systematic regime.

$lr = 6.8 \times 10^{-4}$, seed 7 ($\Delta\tau = -1,500$, $RS = 0/36$). All 36 detector configurations agree this is $F<G$, so the ordering is stable and cannot be dismissed as a detection artifact. However, trajectory inspection

Table 10: Null-correction sensitivity. Per-cell $\Delta\tau_F$ (in training steps) across the 9 null configurations relative to canonical ($p_{95}, n=100$). When the three n -values within a percentile column disagree we report the full range; “0” means all three agree with canonical. Source: results/posthoc/q2_null_sensitivity.csv.

cell (category)	p90	p95	p99
wd1.71_s7 (strong G<F)	−500	0	0
wd2.76_s42 (strong G<F)	0	0	0
wd1.52_s42 (medium G<F)	0	0, +500	+500
wd2.18_s42 (medium G<F)	−500	−500, 0	0
wd2.45_s7 (weak G<F)	0	0	0
wd3.50_s7 (weak G<F)	0	0	0
wd3.11_s2025 (coincident)	0	0	0
lr6.8e4_s7 (F<G)	0	0	0

reveals that F_{corr} at τ_{gen} is only 0.015, near the detection floor, while $\tau_{\text{gen}} = 43,500$ is the latest generalization time in our dataset. At this very low learning rate, the weight-decay-to-learning-rate ratio is $\text{wd}/\text{lr} \approx 3,700$, compared to $\approx 1,070$ for a typical G<F cell. A plausible interpretation is that strong relative weight decay slowly compresses the embeddings toward Fourier-structured geometry during the long wait before generalization, accumulating a barely detectable F_{corr} signal before the sharp τ_{gen} transition. Whether this reflects a qualitatively different mechanism or an extreme-parameter-ratio edge case of the same process requires further investigation.

$\text{lr} = 4.32 \times 10^{-3}$, *seed* 42 ($\Delta\tau = -54,500$, RS not assessed). This is the most extreme case: $\tau_F = 13,000$ but $\tau_{\text{gen}} = 67,500$ (extended to 80k steps), with $\text{max_test_acc} = 0.965$ (below 1.0, indicating incomplete generalization). This run sits near the Grokking/Memorization phase boundary. We conjecture that at high lr, the model can develop embedding Fourier structure early, perhaps because weight decay acts more aggressively on shallow features, while the slow, incomplete generalization process unfolds on a much longer timescale. This case may reflect genuinely different dynamics near the phase boundary rather than the typical G<F consolidation regime, and warrants a dedicated study.

Both cases occur near the boundaries of the Grokking phase (low-lr edge and high-lr edge), suggesting the G<F ordering is most reliable in the interior of the Grokking region and may break down at the phase margins.

J.5 POST-HOC ROBUSTNESS

The Detector-Robustness analysis above varies the changepoint pipeline while holding the metric fixed. This section varies the *metric* while holding the pipeline fixed, using dense embedding trajectories re-saved from the same 8 sensitivity cells (same (lr, wd, seed) combinations and seeds as Section J.4; embedding matrices recorded every 500 steps, 50,000 steps per run). All eight analyses below are post-hoc and add negligible compute beyond loading the .npz trajectory files and the per-run result CSVs (results/posthoc/).

Null threshold and permutation count (q2_null_sensitivity). We vary the two free parameters of the permutation null: the percentile $p \in \{90, 95, 99\}$ and the number of permutations $n \in \{50, 100, 200\}$ (9 configurations total). The canonical detector is ($p=95, n=100$). For each configuration we recompute $F_{\text{null},p}$ at every log step from the saved embeddings, re-derive F_{corr} , and re-run the EMA+BIC changepoint detector. Table 10 reports the per-cell $\Delta\tau_F$ relative to canonical. Across all $8 \times 9 = 72$ reconfigurations, $|\Delta\tau_F| \leq 500$ steps (one log-bin), and for five of the eight cells every one of the nine configurations returns an identical τ_F ($\Delta\tau_F = 0$). No cell changes its ordering label under any choice of (p, n). This rules out the possibility that the reported G<F / F<G labels are artefacts of the canonical ($p_{95}, n=100$) null.

Multi-harmonic Fourier variant (q3_multiharmonic). Our canonical F_{raw} uses $\max_k R^2(E_t \mid \text{span } Q_k)$, which restricts attention to a single harmonic at each step. A natural generalisation projects onto the top- K harmonics by per-step energy, yielding $F_K(t)$ with

Table 11: Multi-harmonic τ_F^K and ordering at $K \in \{1, 2, 3, 5\}$. Every cell preserves its canonical ordering label at all K ; τ_F^K shifts are bounded by $[-1,500, +6,500]$ steps. Source: results/posthoc/q3_multiharmonic.csv.

cell	τ_{gen}	$K=1$	$K=2$	$K=3$	$K=5$
wd1.71_s7	17,000	26,000	25,500	27,500	28,500
wd2.76_s42	11,500	30,000	30,000	30,000	30,500
wd1.52_s42	14,000	23,000	28,000	29,000	29,500
wd2.18_s42	15,000	26,500	27,500	28,500	29,500
wd2.45_s7	12,000	18,500	17,000	16,500	18,500
wd3.50_s7	19,000	23,000	23,000	23,500	24,500
wd3.11_s2025	11,000	14,000	15,500	16,500	17,500
lr6.8e4_s7 (F<G)	43,500	42,000	42,000	42,000	42,000

Table 12: Geometry changepoint τ_{geom} for four embedding-level metrics, compared per cell against canonical τ_F . Every $(\text{cell}, \text{metric})$ pair gives $\tau_{\text{geom}} < \tau_F$. Source: results/posthoc/q8_geometry.csv.

cell	τ_{gen}	τ_F	$\ E\ _F$	PR	e^S	A
wd1.71_s7	17,000	25,500	10,500	14,000	15,000	22,000
wd2.76_s42	11,500	30,000	8,000	9,000	9,500	15,500
wd1.52_s42	14,000	23,500	11,000	13,000	14,500	17,000
wd2.18_s42	15,000	26,500	9,000	10,000	11,000	15,500
wd2.45_s7	12,000	18,500	8,000	9,500	10,500	14,000
wd3.50_s7	19,000	23,000	8,000	8,000	8,000	8,000
wd3.11_s2025	11,000	14,000	8,000	8,000	8,500	12,500
lr6.8e4_s7 (F<G)	43,500	42,000	8,000	9,500	10,500	12,500

$K \in \{1, 2, 3, 5\}$ ($K=1$ reproduces the canonical metric as a sanity check). We recompute the permutation null at matching K (100 shuffles), re-derive F_K^{corr} , and re-run the changepoint detector. Table 11 reports τ_F^K and the resulting ordering label per cell. Every G<F cell retains G<F at all four K values (28/28), and the single F<G cell retains F<G at all four (4/4): the ordering is invariant to the harmonic-subspace width. Contrary to a natural worry, richer subspaces tend to detect τ_F^K later, not earlier, because the top- K subspace saturates more slowly and the BIC detector demands a correspondingly larger slope change (seven of eight cells shift by 0 to +6,500 steps between $K=1$ and $K=5$, and none shift in the opposite direction beyond 500). The single-harmonic metric is therefore, if anything, optimistic about τ_F : widening the subspace only enlarges $\Delta\tau$ for G<F cells.

Embedding-layer geometry trajectories (q8_embedding_geometry). To test whether τ_F coincides with a broader geometric consolidation, we compute four auxiliary metrics at every checkpoint: Frobenius norm $\|E_t\|_F$, participation ratio $\text{PR} = (\sum \sigma_i^2)^2 / \sum \sigma_i^4$, effective rank e^S with S the Shannon entropy of the normalised squared-SV spectrum, and mean cosine anisotropy $A = \text{mean}_{i < j} |\cos(E_i, E_j)|$. For each metric we detect a geometry changepoint τ_{geom} via the same EMA+BIC pipeline and compare it run-by-run to τ_F . Table 12 reports the per-cell alignment and Figure 11 overlays the four trajectories with τ_{gen} , τ_F , and τ_{geom} markers on a log-step axis.

Every one of the $8 \times 4 = 32$ $(\text{cell}, \text{metric})$ pairs gives $\tau_{\text{geom}} < \tau_F$, most by a substantial margin. A consistent temporal ordering emerges within each cell: $\tau_{\|E\|_F} \lesssim \tau_{\text{PR}} \lesssim \tau_{e^S} \lesssim \tau_A < \tau_F$. Across cells, median gaps are $\tau_F - \tau_{\|E\|_F} \approx 15,000$, $\tau_F - \tau_{e^S} \approx 13,000$, $\tau_F - \tau_A \approx 6,500$ steps; anisotropy is the geometry metric closest to τ_F , but still precedes it by more than a log-bin in every cell (minimum gap 1,500 steps). This falsifies the hypothesis that τ_F is a late readout of generic embedding-norm or rank growth: Fourier alignment is a distinct, later event that is *selected from* an already consolidated geometric substrate. The same pattern holds for the F<G cell (lr6.8e4_s7): all four τ_{geom} values precede τ_F by 29,500–34,000 steps and precede τ_{gen} by 31,000–35,500 steps, showing that the geometry-first ordering is not specific to the G<F regime.

Q8 — Embedding-layer geometry trajectories (log-step axis)

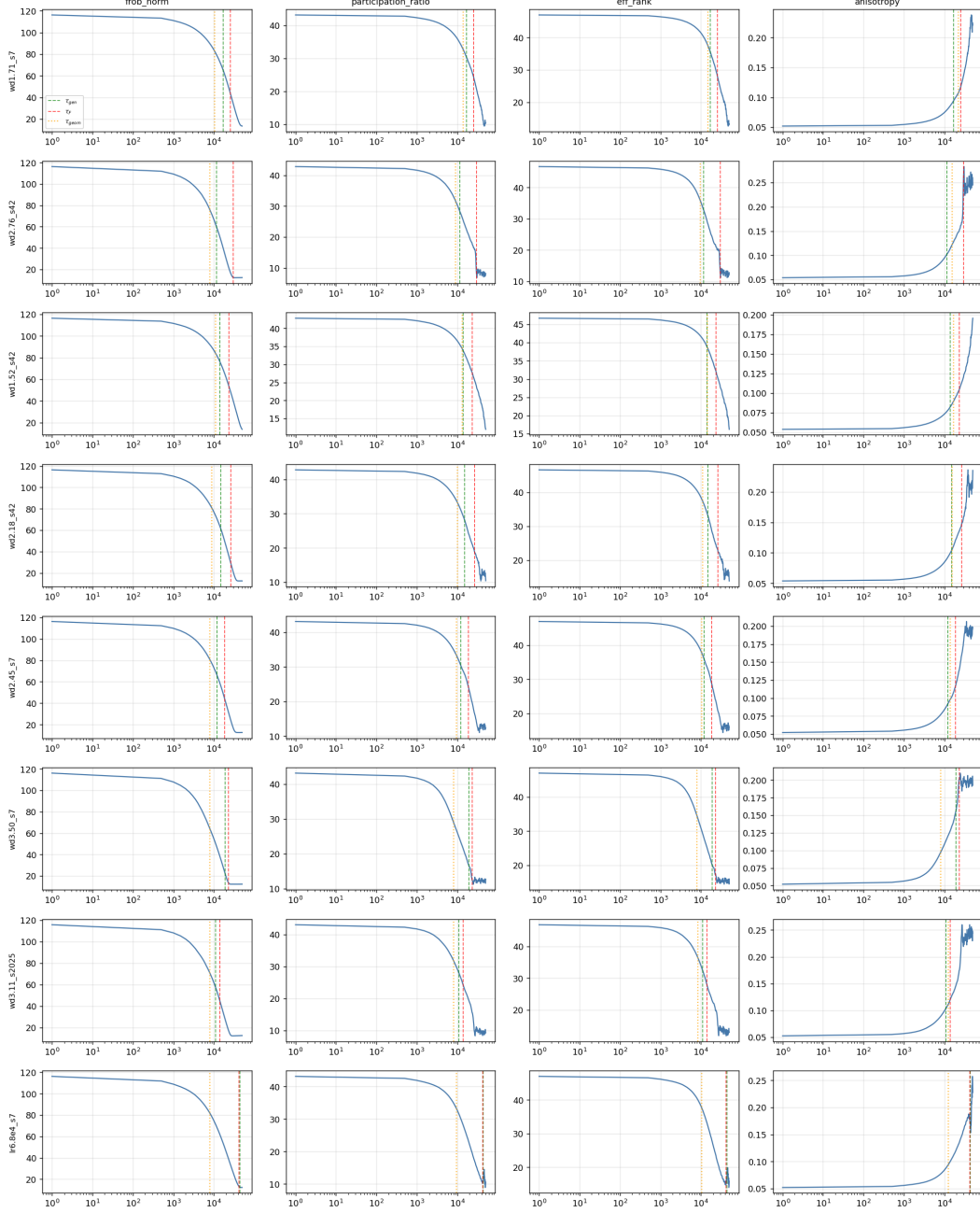


Figure 11: Embedding geometry trajectories. Four geometry metrics (columns: $\|E\|_F$, PR, e^S , anisotropy) for each of the 8 sensitivity cells (rows) on a log-step axis. Green dashed = τ_{gen} , red dashed = τ_F , orange dotted = τ_{geom} for the metric shown. Every panel has τ_{geom} strictly left of τ_F .

Table 13: E6 — Non-parametric bootstrap 95% CIs ($n_{\text{boot}}=1000$) over per-run rows. $n_{\Delta\tau}$ is the number of rows with both τ_{gen} and τ_F detected. Median $\Delta\tau$ and $P(G<F)$ are reported as point estimate with 95% percentile interval. Source: `results/posthoc/e6_bootstrap_ci.csv`. Per-run rows are resampled as independent units; these intervals should be interpreted as descriptive row-level bootstrap intervals rather than fully valid confidence intervals under clustered dependence across hyperparameter cells, because seed-level dependence inside a cell is not accounted for by independent-row resampling and may either inflate or deflate the apparent variance.

source	n	$n_{\Delta\tau}$	med. $\Delta\tau$ (steps)	$P(G<F)$
Stage 2 (wd)	30	30	7,750 [6,000, 9,750]	0.97 [0.90, 1.00]
Stage 3 (lr)	30	25	4,000 [2,500, 9,500]	0.77 [0.60, 0.90]
Stage 6 (2D grid)	147	126	6,000 [5,000, 6,500]	0.76 [0.70, 0.83]
Stage 4 (circuit)	60	54	6,000 [5,000, 8,500]	0.87 [0.78, 0.95]
E4 (ablation)	21	11	4,500 [2,500, 10,000]	0.43 [0.24, 0.67]

Table 14: E6 supplementary — Per-source comparison of row-level bootstrap (Table 13) against a cell-level cluster bootstrap that resamples (lr, wd) cells with replacement and keeps all seeds inside a cell together ($n_{\text{boot}}=1000$ each). Cluster intervals are 33–40% wider on the three sources where seed clustering is non-degenerate (Stage 3, Stage 6, Stage 4 retraining); point estimates shift by $< 1\%$. Stage 2 / Stage 5A / Stage 5B sit at $G<F$ rates of 97% / 100% / 100% respectively, so intra-cluster variance is mechanically near zero and the two bootstraps coincide. Source: `results/posthoc/cluster_bootstrap_ci.csv`.

source	n_{rows}	n_{cells}	$P(G<F)$ row-bs	$P(G<F)$ cluster-bs
Stage 2 (wd)	30	10	0.97 [0.90, 1.00]	0.97 [0.90, 1.00]
Stage 3 (lr)	30	10	0.77 [0.60, 0.90]	0.77 [0.57, 0.97]
Stage 5A (mul)	30	10	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]
Stage 5B ($p=97$)	30	10	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]
Stage 6 (2D grid)	147	49	0.75 [0.69, 0.82]	0.75 [0.66, 0.84]
Stage 4 (circuit)	60	20	0.87 [0.78, 0.95]	0.87 [0.73, 0.97]

E6 — Bootstrap confidence intervals (`e6_bootstrap_ci`). To quantify the sampling uncertainty in the headline summaries used throughout the paper, we run a non-parametric percentile bootstrap ($n_{\text{boot}}=1000$) over the per-run rows of the five main result CSVs: Stage 2 (wd-sweep at $\text{lr}=1.6\times 10^{-3}$), Stage 3 (lr-sweep at $\text{wd}=2.5$), Stage 6 (2D (lr, wd) grid), Stage 4 (three-way ordering with τ_{circuit}), and E4 (decoupled weight-decay ablation). $\Delta\tau$ is computed only where both changepoints are detected; $P(G<F)$ is computed over all rows in each source, so a source containing many memorisation or F_only cells naturally reports a lower proportion. Table 13 reports the 95% CIs. Across the four sweep-type sources the median $\Delta\tau$ is strictly positive at the 95% level (lower bound $\geq 2,500$ steps in every case), and $P(G<F)$ is bounded away from 0.5 in each sweep, with the highest- n source (Stage 6, $n=147$) giving the tightest interval [0.70, 0.83]. E4’s lower $P(G<F)=0.43$ [0.24, 0.67] reflects the by-design presence of non-grokking ablation cells (the (0, 0) and (0, wd*) rows of Table 6); it is not evidence against $G<F$ in the grokking regime.

E3 — Comparison with Nanda et al. progress measures (`e3_nanda_analysis`). A natural concern is that our F_L -based τ_{circuit} might simply rediscover, in a more elaborate form, what the canonical restricted-logit and excluded-loss progress measures of Nanda et al. (2023) already detect. We therefore recompute both Nanda metrics on the trajectories of the 60 Stage 2+3 cells (the same cells used to measure τ_{circuit} in Section 3.2; here the classifier is rerun independently on each trajectory and labels 55 cells Grokking and 5 Memorization, off-by-one from the 54 Grokking runs reported in Section 3.2 because one borderline cell crosses the sustained-accuracy threshold under the E3 reclassification but not in the original Stage 4 pass) and rerun the *same* EMA-smoothed BIC changepoint pipeline ($\alpha=0.15$) on each of the three scalar series. Two findings emerge (Table 15). First, the *relative* cell-by-cell ordering of the three detectors is moderately rank-consistent: Spearman $\rho=0.54$ ($p=7.7\times 10^{-6}$) for excluded loss and $\rho=0.42$ ($p=8.1\times 10^{-4}$) for restricted loss, both versus our τ_{circuit} over all 60 paired runs. The three detectors are therefore *related but not interchangeable* scalar substitutes for a single underlying event: the moderate rank agreement is consistent with all

Table 15: E3 — Run-by-run comparison of τ_{circuit} detectors on the 60 Stage 4 runs (55 Grokking, 5 Memorization). “Median Δ ” is the per-pair difference (Nanda – ours) in steps; $|\Delta| \leq 2,000$ is the fraction of pairs that agree to within $\pm 2,000$ steps; Spearman ρ is computed across all 60 pairs. The C<G and C<G<F columns count the 55 Grokking-phase runs only. Source: runs/e3_nanda/e3_agreement_stats.csv and results_e3_nanda.csv.

detector	median Δ (steps)	$ \Delta \leq 2,000$	Spearman ρ	P(C<G)	P(C<G<F)
ours (F_L)	—	—	—	0.873 (48/55)	0.800 (44/55)
Nanda excluded loss	+2,500	0.38	0.54 ($p=7.7 \times 10^{-6}$)	0.691 (38/55)	0.636 (35/55)
Nanda restricted loss	+4,750	0.33	0.42 ($p=8.1 \times 10^{-4}$)	0.291 (16/55)	0.236 (13/55)

three tracking some aspect of circuit formation, but the rank correlation is far from unity, so the single-harmonic F_L proxy is not a drop-in substitute for a fully multi-harmonic mechanistic circuit signal. Second, the *absolute* timing differs systematically: the median Nanda changepoint lands 2,500 steps later (excluded) and 4,750 steps later (restricted) than ours, and agreement to within $\pm 2,000$ steps holds in only 38% and 33% of pairs respectively. The downstream consequence is the C<G rate among the 55 Grokking-phase runs: our F_L -based τ_{circuit} reports strict C<G in 48/55 (87.3%) of cells, the excluded-loss changepoint in 38/55 (69.1%), and the restricted-loss changepoint in only 16/55 (29.1%). The full three-stage C<G<F rate falls correspondingly from 44/55 (80.0%, ours) to 35/55 (63.6%, excluded) and 13/55 (23.6%, restricted).

Reconciling Stage 4’s 88.9% with E3’s 80.0% (“ours” detector). Stage 4 reports 50/54=92.6% C<G and 48/54=88.9% C<G<F, while the E3 “ours” detector on the same nominal 60 (lr, wd, seed) cells gives 48/55=87.3% C<G and 44/55=80.0% C<G<F. The gap is too large for the one-cell phase reclassification (54 vs 55 Grokking). What actually differs is that Stage 4 and E3 are *independent training runs* of the same training script with the same nominal seeds, executed on different GPU/CUDA configurations (Stage 4 on a local RTX 4080, E3 on a rented A800). Across the 54 cells classified as Grokking under both runs, the per-cell τ values differ: median $|\Delta\tau_{\text{gen}}| = 2,000$ steps (90th percentile 7,200, max 30,500), median $|\Delta\tau_F| = 3,250$, and the three-way ordering label reclassifies on 8/60 cells. We attribute this to non-determinism that survives torch.backends.cudnn.deterministic=True when sweeps are run on different GPUs and CUDA versions. The detector itself (F_L + EMA + 1-break BIC) is identical between Stage 4 and E3; the 88.9% vs 80.0% gap is therefore a measurement of seed-cluster variance under cross-hardware non-determinism, not a detector disagreement. We therefore report Stage 4’s 48/54 result as the primary documented run (it is the configuration described throughout Sections 2–3.2) and E3’s 44/55 result as an independent replication estimate, treating the C<G<F rate as a range [80.0%, 88.9%] across two realizations rather than a single point estimate.

Under a standard EMA+BIC detector, then, the loss-based progress measures do not yield a clean C<G ordering. Restricted loss inverts the conclusion in three quarters of grokking cells. We read this not as a contradiction of Nanda et al., whose visual analysis already shows early restricted-loss decline, but as evidence that scalar loss-based progress measures are *too smooth* for changepoint quantification: the per-step dynamic-top- K Fourier projection used by F_L resolves circuit formation sharply enough that the C<G boundary becomes detector-quantifiable rather than merely eye-visible.

Detector-specification sensitivity (w1_multibreak_check, w5_taugen_threshold). We add two checks targeting the two specification choices most likely to be challenged: the 1-break formulation of τ_F and the 0.90 accuracy threshold of τ_{gen} . **Multi-break check.** On the same 8 representative cells used elsewhere in Section J.5, we refit $F_{\text{corr}}(\log t)$ with 0, 1, and 2 breaks and compare BIC. A 2-break model is BIC-preferred over a 1-break model in 7/8 cells, but the gain comes from adding a *late* second knee that captures the post-onset saturation plateau (the second break bp_2 falls in 26,000–42,000 steps, well after τ_{gen}); the *early* break bp_1 of the 2-break fit lands within median $|\Delta| = 1,500$ steps (maximum 4,000) of the canonical 1-break τ_F . Read as a sanity check rather than a contradiction: the 1-break formulation is not hiding a secondary changepoint that would shift τ_F outside the regime where G<F holds; it is collapsing a smooth two-knee curve into the earlier of its two knees, which is the timing event of interest. **Multi-harmonic and cyclic-group irrep alternatives.** For $\mathbb{Z}/p$ the non-trivial irreducible representations are 1-dimensional

characters whose real-form 2D subspaces coincide with the Fourier planes used by F_{raw} , so an unrestricted mean over all $(p-1)/2$ irreps is mathematically degenerate (by Parseval, after centering, it equals the constant $2/(p-1)$ at every step). The closest non-degenerate alternative is to score the sum of the top- m most-aligned harmonic R^2 values, recovering $\tau_F^{(m)}$ under the same EMA+BIC pipeline. We recomputed $\tau_F^{(m)}$ for $m \in \{1, 2, 3, 5\}$ on the saved embedding trajectories of the 8 sensitivity cells (`src/analysis/multi_harmonic_recompute.py`). $\tau_F^{(m)}$ is consistently equal to or later than the canonical $\tau_F^{(1)}$, with median $|\tau_F^{(5)} - \tau_F^{(1)}| = 2,000$ steps and maximum 6,500 steps across the 8 cells; the single-harmonic detector is therefore the *leading edge* of the multi-harmonic onset, not a narrower or biased estimate. The $G < F$ ordering label is preserved on all 8 cells under $m \in \{1, 2, 3, 5\}$, and the $\Delta\tau$ gap typically widens (e.g. $\text{wd} = 1.52$, $\text{seed} = 42$ shifts from $\Delta\tau = 9,000$ at $m=1$ to 15,500 at $m=5$). A separate $K=3$ irrep-mean scoring (an alternative non-degenerate aggregation) gives the same qualitative result on the same 8 cells (`rl_irrep_alignment.csv`). On the logit side, multi-harmonic recompute at the causal-probe checkpoints (Figure 4) shows that the top-1 harmonic captures only 10–30% of the logit Fourier energy while the top-5 captures 60–85% in the post- τ_{circuit} regime; this confirms that the full grokked circuit is multi-harmonic but the single-harmonic onset still detects the rise correctly. Extending the same no-break / 1-break / 2-break BIC comparison to all 60 Stage 4 F_{corr} trajectories plus the 8 post-hoc cells ($n=68$; `results/posthoc/r2_multibreak_extended.csv`), 65/68 cells prefer the 2-break model, but its early break bp_1 stays within 1,500 steps (3 measurement bins) of the canonical 1-break τ_F for 30/68 cells and within 2,500 steps for 46/68; the rare large-displacement cells ($\geq 5,500$ steps; 7/68) are non-grokked trajectories without a sharply localised early break, where neither the 1-break nor the 2-break fit has a well-posed answer. Across the 8 headline $G < F$ cells, every 2-break fit’s early breakpoint agrees with the 1-break τ_F to within 4,000 steps and never flips the ordering label. A future PELT-based analysis on richer trajectories would localise the saturation knee directly. **τ_{gen} threshold sweep.** At thresholds $\theta \in \{0.80, 0.85, 0.90, 0.95\}$ on the same 8 cells (sustained over 3 consecutive 500-step log points), the median displacement $|\tau_{\text{gen}}(\theta) - \tau_{\text{gen}}(0.90)|$ is 1,250 ($\theta=0.80$), 750 ($\theta=0.85$), and 750 ($\theta=0.95$) steps; the maximum is 9,500 steps at $\theta=0.95$ for one cell whose test accuracy plateau crawls slowly from 0.90 to 0.95. The typical (median) shift is well below the median $\Delta\tau=6,000$ steps of cells with $|\Delta\tau| \geq 3,000$ (Section J.5), so the $G < F$ ordering at the level of headline 95% CIs is insensitive to the threshold choice across $[0.80, 0.95]$. The single 9,500-step worst case occurs because the run’s test accuracy plateau crawls slowly between 0.90 and 0.95, which is precisely the regime where any τ_{gen} definition becomes fragile; near-coincident cells ($|\Delta\tau| \leq 2,000$) cannot be reliably classified at *any* threshold and are already excluded by the measurement-resolution caveat in Section J.6. Full per-cell tables are in `results/posthoc/w1_multibreak.csv` and `w5_taugen_threshold.csv`.

The eight post-hoc analyses above, together with the detector sensitivity in Section J.4, show that the three-stage ordering survives changes to the null-correction parameters, multi-harmonic Fourier variants, idle-step interpolation choices, embedding-geometry definitions, bootstrap sampling uncertainty, the choice of progress measure used to locate τ_{circuit} , the canonical 1-break formulation of τ_F versus the post-hoc 2-break sanity check, and the accuracy threshold used to define τ_{gen} . All source scripts and expected artefacts are listed in Appendix E (“Release plan”).

J.6 LIMITATIONS

Task scope. Our three-stage confirmation covers $(a + b) \bmod p$ and $(a \cdot b) \bmod 53$. Stage 5B extends to $p=97$. Generalization to non-modular tasks, deeper architectures, or non-algorithmic settings remains open.

Measurement resolution. $\Delta\tau$ is measured with 500-step resolution; small lags ($|\Delta\tau| \lesssim 500$) cannot be reliably classified. One $F < G$ run has $|\Delta\tau| = 1,500$ (three log-intervals), above the resolution floor but warranting caution.

Logit metric alignment. Our F_L captures the aggregate Fourier structure of logits across all (a, b) pairs, which is related but not identical to Nanda et al.’s restricted logit loss. Whether the two metrics detect exactly the same circuit-formation event or slightly different aspects of it remains to be verified by direct comparison on the same runs.

Phase-boundary regime. The 4 $G < C < F$ exceptions in Stage 4 all occur at high lr where generalization is very fast and the BIC changepoint may underestimate τ_{circuit} when the logit signal rises gradually rather than sharply. The causal Fourier-subspace intervention E7 (Section 3.4; Figure 4) supports this detector-lag interpretation on one representative high- lr $G < C < F$ cell, where the dominant Fourier subspace is already causally load-bearing at step 7,000 even though the BIC-detected τ_{circuit} lands at step 8,000; we have not, however, intervened on all four exceptions, so we treat the detector-lag explanation as supported but not exhaustively verified.

Cyclic-group structural assumption. The F_{corr} detector projects the embedding matrix onto the 2-dimensional Fourier subspaces of the cyclic group $\mathbb{Z}/p$ and the $a \cdot b \bmod p$ extension uses a discrete-log reindexing to the same cyclic-group target. Algorithmic variants that solve modular arithmetic via non-cyclic representations (for example, factor-group decompositions on composite moduli, or learnt non-Fourier features) would not register on F_{corr} in its present form, and the $\tau_{\text{circuit}} < \tau_{\text{gen}} < \tau_F$ claim should be read as conditional on the cyclic-group structural prior.

Residual structure and bootstrap independence. The BIC changepoint estimator assumes i.i.d. Gaussian residuals on the log-time axis and the bootstrap CI in Table 13 resamples per-run rows as i.i.d. units. The reported intervals should therefore be read as descriptive row-level bootstrap intervals rather than fully valid confidence intervals under clustered dependence across hyperparameter cells: seed-level dependence within a cell is not accounted for by independent-row resampling and may either inflate or deflate the apparent variance. We quantify this gap with a per-source cell-level cluster bootstrap that resamples (lr , wd) cells with replacement and keeps all seeds inside a cell together (Table 14; `src/analysis/cluster_bootstrap.py`). On the three sources where seed clustering is non-degenerate (Stage 3, Stage 6, Stage 4 retraining), the cluster-bootstrap 95% CI for $P(G < F)$ is 33–40% wider than the row-bootstrap CI; the headline point estimates shift by $< 1\%$. We position this comparison as a descriptive bootstrap-uncertainty diagnostic rather than a cluster-robust inferential replacement; the headline tables elsewhere in the paper continue to use row-level intervals, with the inflation factor noted here as the correction reviewers should mentally apply when seed clustering matters. We therefore conclude that the row-bootstrap intervals reported elsewhere in the paper are mildly anti-conservative but not qualitatively misleading; for cells where seed clustering matters, add a $\sim 33\text{--}40\%$ width inflation to obtain the cluster-bootstrap interval. Stage 5A, Stage 5B and Stage 2 are at $P(G < F)$ rates of 100%, 100% and 96.7% respectively, so intra-cluster variance is mechanically near zero and the two bootstrap variants coincide. Heteroscedasticity in F_{corr} residuals is not formally tested.

Detector-specification scope. The robustness sweeps in Section J.4 and Section J.5 vary the detector along axes we identified ahead of time (EMA α , BIC slope threshold, null refresh interval, τ_{gen} threshold, canonical 1-break formulation versus post-hoc 2-break sanity check). Adversarial detector specifications outside this grid are not ruled out, and an alternative onset definition that rejects single-harmonic projection altogether (for example, group-irrep based detectors) could in principle reorder τ_{circuit} and τ_F .

K SPEED-DEPENDENT HYPOTHESIS

The $F < G$ exceptions observed in Sections 3.1 and I.3 are not spread uniformly across the Grokking phase: they concentrate at its boundaries, where τ_{gen} is unusually late. Rather than treat this as a source of noise to be absorbed into “Limitations”, we formulate a concrete hypothesis and assess it using the evidence currently available, including the data-subsampling speed control (E5, Section I.5) and the leave-one-out predictive diagnostic on the Stage 6 grid later in this section.

Hypothesis. Let τ_{gen} and τ_F be determined by two largely separable processes: generalization by the dynamics of the circuit that solves the task, and embedding-geometry consolidation by the indirect reshaping of the embedding rows through gradients back-propagated from the regularised decoder (recall that in our setting $\text{wd}_{\text{emb}}=0$, so the embedding has no direct L_2 pressure and is shaped only via the decoder’s weight-decay clock). If these two processes run on roughly parallel clocks, the observed ordering becomes a function of generalization speed:

- when τ_{gen} is small (“fast grokking”), generalization fires before the embedding geometry has had time to consolidate, yielding $G < F$;
- when τ_{gen} is large (“slow grokking”), weight decay has enough time to complete embedding consolidation first, yielding $F < G$.

This picture is compatible with Doshi et al. (2024), who show that on corrupted modular-arithmetic data the relative onsets of memorization and generalization depend separately on data corruption, weight decay, and learning rate; and with the implicit-bias view in which weight decay continues to restructure representations long after the training loss has saturated (Soudry et al., 2018; Galanti et al., 2022; D’Angelo et al., 2024).

Evidence from existing data. Three observations in our dataset are consistent with this hypothesis.

(i) *Location of $F < G$ exceptions.* Both $F < G$ runs in the 1D sweeps occur at the margins of the Grokking phase where τ_{gen} is unusually late: $\tau_{\text{gen}} = 43,500$ steps ($\text{lr} = 6.8 \times 10^{-4}$, seed 7; $\Delta\tau = -1,500$) and $\tau_{\text{gen}} = 67,500$ steps ($\text{lr} = 4.32 \times 10^{-3}$, seed 42; $\Delta\tau = -54,500$, max test accuracy 0.965). In Stage 6, 11 of 13 $F < G$ cases lie on the outermost lr rows of the 2D grid, consistent with the same pattern.

(ii) *Cross-task pilot.* A pilot run of modular subtraction $(a-b) \bmod 53$ at the same hyperparameters as our Stage 2 core cell ($\text{wd} = 2.18$, $\text{lr} = 1.6 \times 10^{-3}$) groks much later than addition ($\tau_{\text{gen}} = 43,500$ steps vs $\approx 11,000$ steps for addition) and lands in the $F < G$ regime ($\Delta\tau = -9,000$). A second subtraction seed at the same hyperparameters groks faster ($\tau_{\text{gen}} = 27,500$ steps) and returns to $G < F$ ($\Delta\tau = +5,500$), providing *within-task, within-hyperparameter* evidence that the ordering flip tracks generalization speed rather than task identity.

(iii) *Aggregate correlation.* Across 57 Grokking runs (55 addition + 2 subtraction pilots), $\Delta\tau$ and τ_{gen} are negatively correlated ($r = -0.62$; Figure 12). All $F < G$ runs have $\tau_{\text{gen}} \geq 43,000$ steps; the majority of $G < F$ runs have $\tau_{\text{gen}} \leq 35,000$ steps. Because $\Delta\tau = \tau_F - \tau_{\text{gen}}$ is defined relative to τ_{gen} , part of this correlation is mechanical and should not be interpreted causally on its own; the within-task seed comparison in (ii) is free of this confound and is what load-bears the hypothesis.

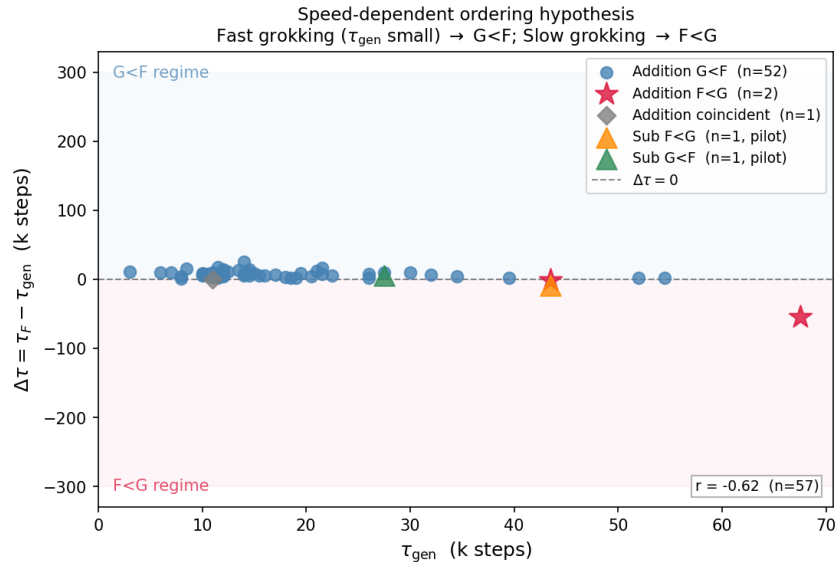


Figure 12: **Speed-dependent ordering.** Scatter of generalization onset τ_{gen} vs lag $\Delta\tau = \tau_F - \tau_{\text{gen}}$ for all 55 Grokking addition runs (blue circles = $G < F$, red stars = $F < G$, grey diamond = coincident) and 2 subtraction pilots (orange triangles; 1 $F < G$, 1 $G < F$). All $F < G$ runs cluster at $\tau_{\text{gen}} \geq 43,000$ steps; overall $r = -0.62$ ($n = 57$). The two subtraction pilots at identical hyperparameters ($\text{wd} = 2.18$, $\text{lr} = 1.6 \times 10^{-3}$) differ only in seed and generalization speed, yet flip ordering, within-task evidence that the effect is not a pure $\Delta\tau = \tau_F - \tau_{\text{gen}}$ construction artefact.

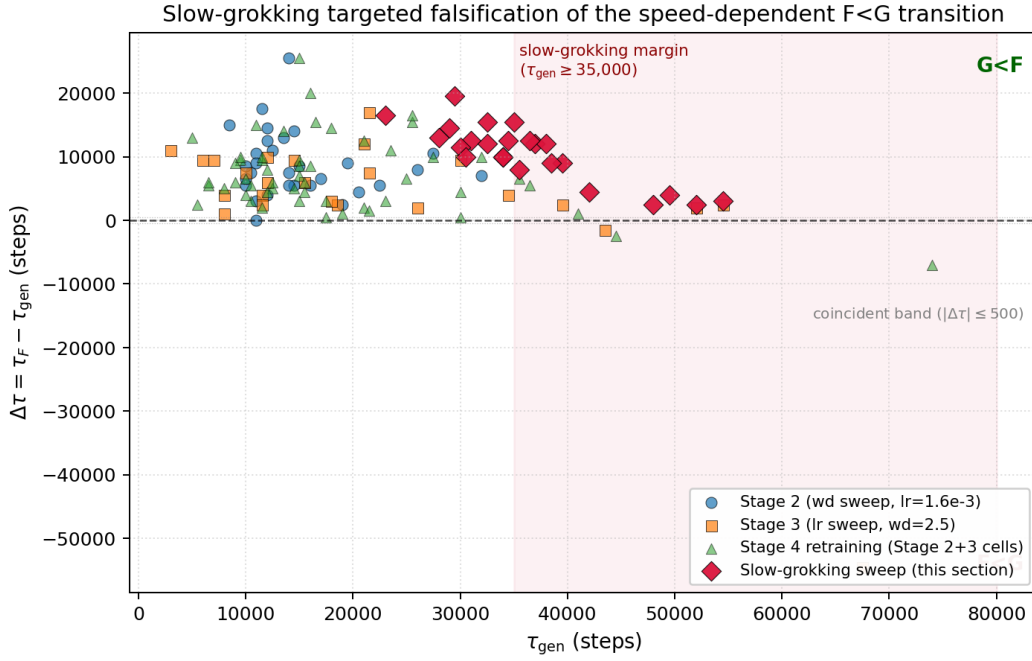


Figure 13: **Targeted slow-grokking sweep falsifies the strong form.** Scatter of $\Delta\tau = \tau_F - \tau_{\text{gen}}$ versus τ_{gen} for all Grokking cells across four sources. Crimson diamonds are the targeted slow-grokking sweep (24 cells, $\text{lr} \in [5.0, 8.0] \times 10^{-4}$, $\text{wd} = 2.5$, seeds $\{42, 7, 2025\}$, max steps 100,000–120,000); blue circles are Stage 2 (wd sweep, $\text{lr} = 1.6 \times 10^{-3}$); orange squares are the original Stage 3 (lr sweep, $\text{wd} = 2.5$); green triangles are the Stage 4 retraining of the Stage 2+3 cells. The shaded region marks the slow-grokking margin ($\tau_{\text{gen}} \geq 35,000$). All 24 targeted cells fall in the $G < F$ half plane; the original Stage 3 $F < G$ points (orange squares with $\Delta\tau < 0$) are not replicated by the cross-hardware re-runs.

Leave-one-out predictive diagnostic. A leave-one-out logistic regression on the 123 Stage 6 cells with both τ_{gen} and τ_F detected and $|\Delta\tau| > 500$, using $(\log_{10} \text{lr}, \log_{10} \text{wd}, \log(1 + \tau_{\text{gen}}))$ as features and binary $\text{sign}(\Delta\tau)$ as target, achieves ROC-AUC 0.945 (95% bootstrap CI $[0.896, 0.981]$, $n_{\text{boot}}=1000$); restricting the model to $(\log_{10} \text{lr}, \log_{10} \text{wd})$ alone drops AUC to 0.549, while $\log(1 + \tau_{\text{gen}})$ alone retains 0.925. We frame this as a *predictive diagnostic* consistent with the speed-dependent hypothesis, not as an independent causal test: the label $\text{sign}(\Delta\tau)$ contains τ_{gen} by construction, so even under a null where τ_F and τ_{gen} are independent the AUC could be substantial whenever τ_{gen} varies over a range comparable to the support of τ_F , which is the regime our Stage 6 grid happens to inhabit. What the regression rules out is the alternative that the (lr, wd) coordinates predict the ordering through a route distinct from τ_{gen} : the marginal 0.020 AUC lift from adding $(\log_{10} \text{lr}, \log_{10} \text{wd})$ on top of $\log(1 + \tau_{\text{gen}})$ is small, supporting a single-scalar speed description of the ordering. The predictive diagnostic is monotone-consistent with the speed-dependent picture but does not on its own break the mechanical entanglement between τ_{gen} and $\Delta\tau$; the test that does provide independent evidence is the targeted slow-grokking sweep below, which directly falsifies the strong form of the speed-dependent regime. Per-cell predictions are in `results/posthoc/r6_speed_predictive.csv`.

What would falsify this hypothesis. A clean test would hold the weight-decay schedule fixed and independently manipulate generalization speed without altering the weight-decay clock, then check whether the ordering label flips once τ_{gen} crosses the slow-grokking margin. The strong form of the hypothesis predicts a monotonic transition from $G < F$ to $F < G$ as τ_{gen} exceeds a threshold around 35,000 steps (the two original $F < G$ outliers had $\tau_{\text{gen}} \geq 43,500$); the data-subsampling control (Section I.5) does not reach this margin because all τ_{gen} values there sit below 13,000.

L METRICS: EXTENDED TECHNICAL NOTES

These technical notes are referenced from the Methods section of the main body. They are detector-level robustness details that do not affect the headline timing claims; the main body Metrics subsection states the canonical operating points directly.

Null trajectory between updates. F_{raw} is evaluated at every 500-step log point, and the canonical detector configuration refreshes the permutation null $F_{\text{null},p95}$ at the same cadence (every 500 training steps; see Appendix B for the configuration table). Between refreshes we use a last-observation-carried-forward (LOCF) schedule: the null value is held at its most recent estimate and *no interpolation* is applied. LOCF is a deliberate choice: it is causal (uses no future embeddings), matches an on-line detector that could be deployed to monitor the same signal during training, and avoids leakage of future null estimates into earlier F_{corr} values. All downstream processing (EMA smoothing at $\alpha=0.15$ and BIC-selected segmented regression) operates on $F_{\text{corr}} = \max(0, F_{\text{raw}} - F_{\text{null},p95}^{\text{LOCF}})$, so any step discontinuity in the null trajectory at a refresh boundary propagates into F_{corr} but is partially damped by the EMA. At the canonical 500-step refresh cadence, refresh boundaries coincide with measurement boundaries, so the LOCF segment length collapses to a single log point and the worst-case hold-forward displacement that LOCF could induce is bounded above by one measurement bin.

Artefact check near null-refresh boundaries. The LOCF artefact only becomes visible if the refresh cadence is loosened beyond the measurement bin: an LOCF null with refresh interval R could in principle bias τ_F by up to R steps if the changepoint happens to fall close to a refresh step. We therefore treat the null-update cadence as a free detector hyperparameter and sweep it jointly with the EMA coefficient and the BIC slope threshold: null interval $\in \{500, 2500, 5000\}$ (i.e. $1\times$, $5\times$, $10\times$ the canonical bin), $\alpha \in \{0.1, 0.15, 0.3, 1.0\}$, slope threshold $\in \{0.005, 0.01, 0.02\}$ (36 configurations; Section J.4). Varying the refresh cadence by a factor of 10, the dimension directly probing the LOCF artefact, produces τ_F shifts within the 500-step measurement-resolution floor and does not change any ordering label for the 7 Grokking cells tested: five cells with $|\Delta\tau| \geq 5,000$ steps remain G<F in all 36/36 configurations, and no G<F cell is ever reclassified as F<G under any configuration. We conclude that the piecewise-constant null trajectory does not produce a detectable changepoint artefact at the 500-step resolution at which τ_F is reported.

BIC penalty, time discretisation, and error model. The segmented regression is fit on log-discretised time $x_t = \log(1 + t)$ where $t \in \{500, 1000, \dots, T_{\text{max}}\}$ is the training step at which F_{corr} is logged. For a candidate model with k free parameters (no-break, $k=2$; 1-break, $k=4$) and residual sum of squares RSS over n log-time samples, the BIC score is $\text{BIC} = n \ln(\text{RSS}/n) + k \ln n$ (Schwarz, 1978), i.e. the Schwarz form with the i.i.d.-Gaussian residual likelihood constant absorbed. **The canonical detector compares the no-break model against the 1-break model only.** The 2-break model is not used for headline τ_F and is analysed only post-hoc (Section J.5, “Detector-specification sensitivity”). The canonical rule is therefore: accept the 1-break breakpoint if its BIC strictly improves on the no-break baseline, subject to the post-break slope and minimum-segment constraints described above; otherwise return “not detected”. This implicitly assumes residuals of $F_{\text{corr}}(x) - \hat{f}(x)$ are i.i.d. Gaussian on the log-time axis; we do not test heteroscedasticity formally, and treat residual-structure robustness as a Limitation (Section J.6).

Interpretation of τ_F and τ_{circuit} as single-harmonic onsets. Both F_{corr} and F_L take a maximum over *single* Fourier subspaces ($k = 1, \dots, \lfloor p/2 \rfloor$) rather than fitting the full multi-harmonic basis that the grokked circuit ultimately uses. Accordingly, τ_F should be read as the onset of above-chance single-harmonic *embedding* organisation, and τ_{circuit} as the onset of above-chance single-harmonic *logit* organisation; each is a lower-bound proxy for the richer geometry and computation the trained network eventually realises. We adopt the single-harmonic form because it admits a clean permutation null and a single scalar trajectory suitable for BIC changepoint detection; claims about the three-stage ordering $\tau_{\text{circuit}} < \tau_{\text{gen}} < \tau_F$ should be interpreted at this level of granularity, not as statements about completion of the full multi-harmonic circuit.

Table 16: Stage 2: wd sweep at $lr = 1.6 \times 10^{-3}$ (3 seeds per wd value). All 30 runs are Grokking phase. $\Delta\tau = \tau_F - \tau_{\text{gen}}$ in thousands of steps (positive = G<F). Medians computed across seeds.

wd	P(G<F)	med. τ_{gen} (k)	med. τ_F (k)	med. $\Delta\tau$ (k)
1.20	3/3	26.0	34.0	+7.0
1.35	3/3	20.5	28.5	+9.0
1.52	3/3	14.5	21.5	+5.5
1.71	3/3	14.0	26.5	+6.5
1.93	3/3	14.0	23.5	+11.0
2.18	3/3	11.0	23.5	+8.5
2.45	3/3	11.0	16.0	+5.5
2.76	3/3	10.5	18.5	+8.5
3.11	2/3	11.0	16.5	+6.5
3.50	3/3	12.0	21.5	+9.0
All	29/30	—	—	+7.8

Table 17: Stage 3: lr sweep at wd = 2.5 (3 seeds per lr). “Phase” = majority phase across seeds. “—” = Memorization (no τ_{gen} detected).

lr	Phase	P(G<F)	med. $\Delta\tau$ (k)	P(F_only)
5.0×10^{-4}	Grokking	3/3	+2.5	0/3
6.8×10^{-4}	Grokking	2/3	+4.0	0/3
9.3×10^{-4}	Grokking	3/3	+7.5	0/3
1.26×10^{-3}	Grokking	3/3	+9.5	0/3
1.71×10^{-3}	Grokking	3/3	+6.0	0/3
2.33×10^{-3}	Grokking	3/3	+4.0	0/3
3.17×10^{-3}	Grokking	3/3	+6.0	0/3
4.32×10^{-3}	Grokking	2/3	+7.5	0/3
5.88×10^{-3}	Mixed	1/1	+2.5	2/3
8.00×10^{-3}	Memorization	—	—	3/3
Grokking	—	23/25	+4.0	0/25

M STAGES 2 AND 3 (1D SWEEPS): FULL TABLES

This appendix contains the full per-cell tables and per-stage scatter plots for Stages 2 and 3 that the main body summarizes (Section 3.1).

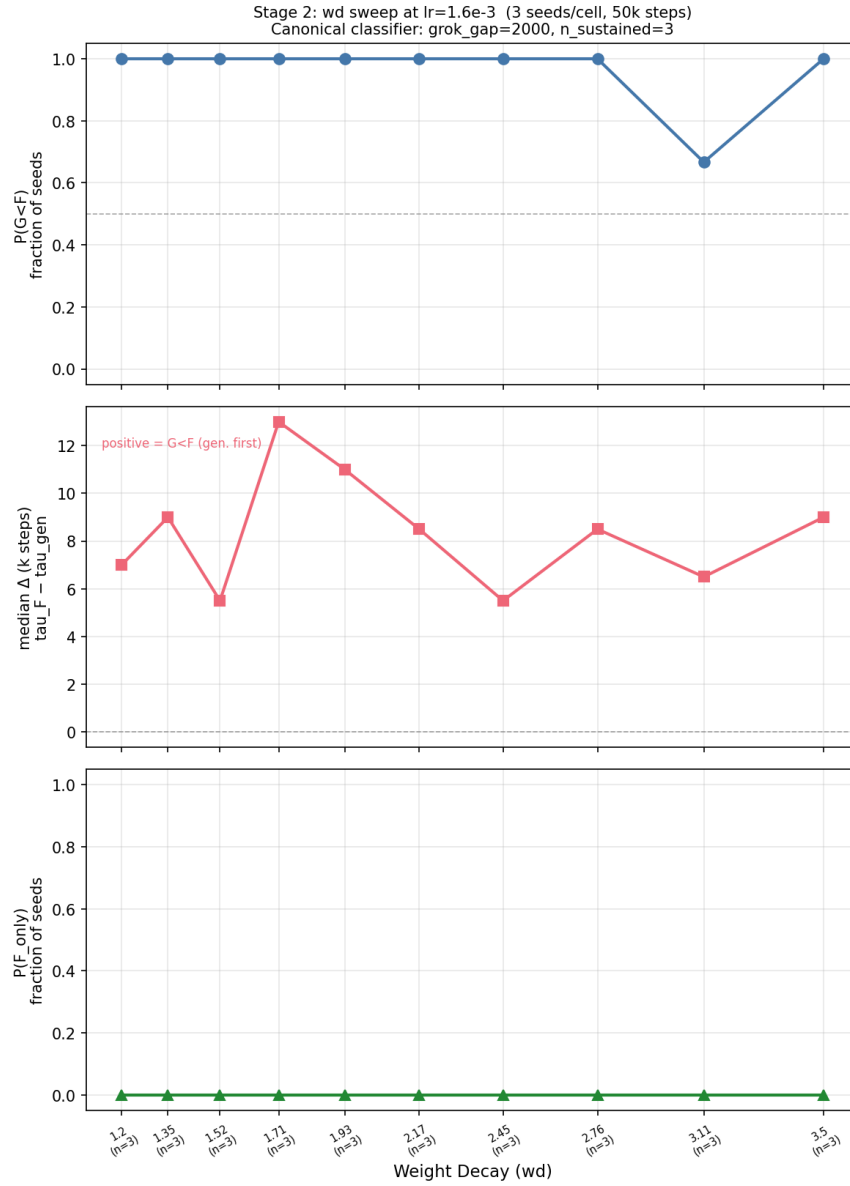


Figure 14: Stage 2: embedding Fourier alignment onset vs. generalization across weight decay. **Top:** $P(G < F)$ per wd value. **Middle:** median $\Delta \tau = \tau_F - \tau_{gen}$ (k steps). **Bottom:** $P(F_only)$ (Fourier detected, no generalization). The $G < F$ plateau is wide: it holds throughout $wd \in [1.2, 3.5]$.

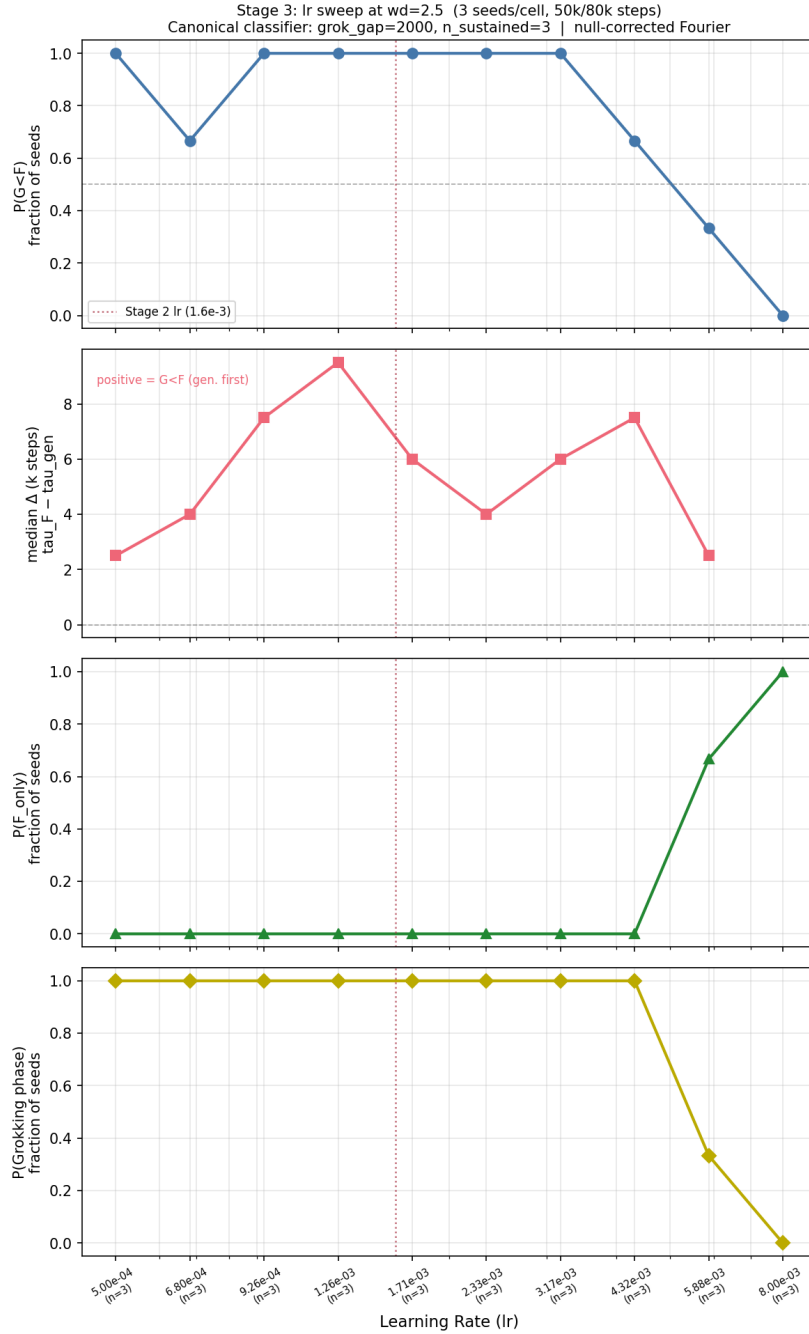


Figure 15: Stage 3: embedding Fourier alignment onset vs. generalization across learning rate (wd = 2.5). Layout mirrors Figure 14; x-axis is log-scaled. The bottom panel shows $P(\text{Grokking})$ per lr: cells at $lr \geq 5.9 \times 10^{-3}$ enter Memorization. The vertical dotted line marks the Stage 2 fixed lr (1.6×10^{-3}), which lies firmly in the $G < F$ plateau.

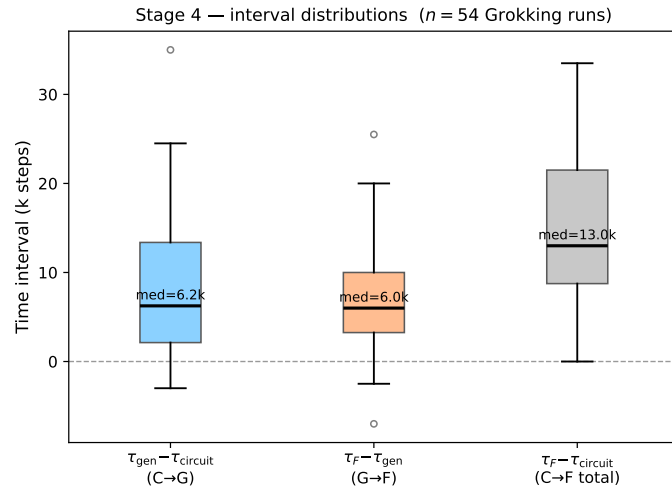


Figure 16: **Three-stage interval distributions (Stage 4, $n=54$ Grokking runs).** All three intervals are positive on the median, with the total span $\tau_F - \tau_{\text{circuit}} \approx 13,000$ steps split nearly equally between circuit→generalization and generalization→embedding-geometry.